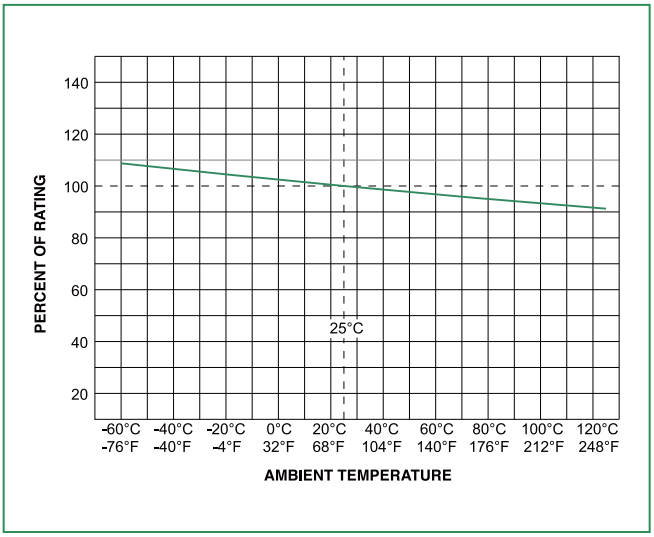
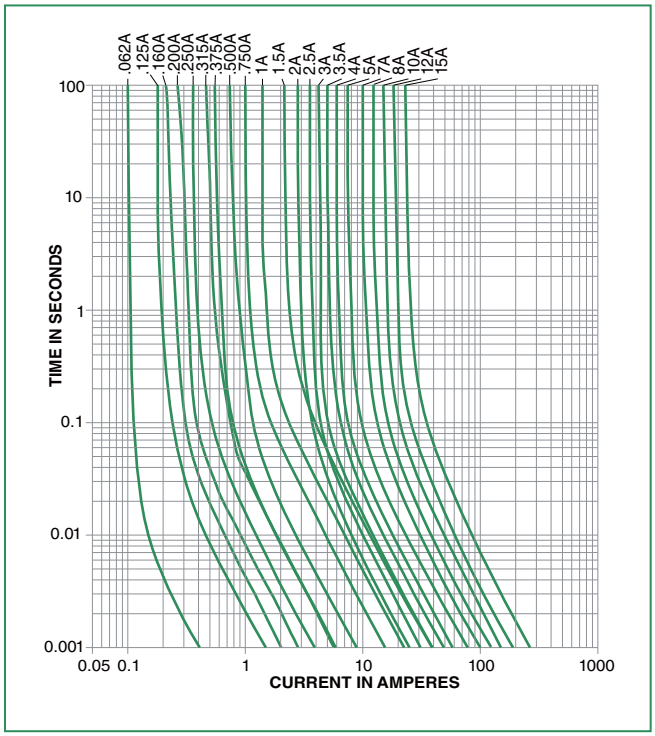
Temperature Rerating Curve



Note:

1. Derating depicted in this curve is in addition to the standard derating of 25% for continuous operation.

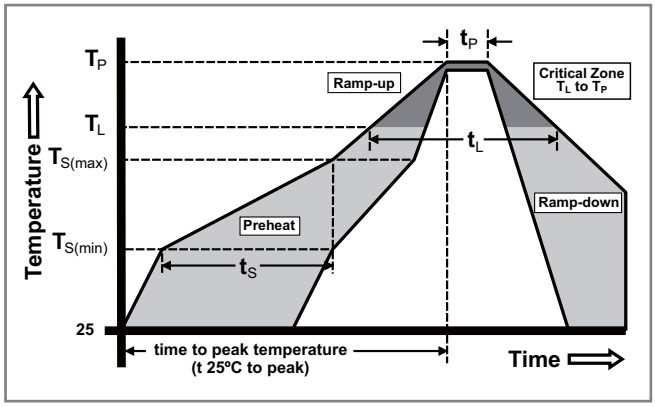
Average Time Current Curves



Soldering Parameters

Reflow Condition		Pb – Free assembly
Pre Heat	-Temperature Min ($T_{s(min)}$)	150°C
	-Temperature Max ($T_{s(max)}$)	200°C
	-Time (Min to Max) (t_s)	60 – 120 secs
Average ramp up rate (Liquidus Temp (T_L) to peak		5°C/second max.
$T_{S(max)}$ to T_L - Ramp-up Rate		5°C/second max.
Reflow	-Temperature (T_L) (Liquidus)	217°C
	-Temperature (t_L)	60 – 90 seconds
Peak Temperature (T_P)		250$^{+0/-5}$ °C
Time within 5°C of actual peak Temperature (t_p)		20 – 40 seconds
Ramp-down Rate		5°C/second max.
Time 25°C to peak Temperature (T_P)		8 minutes max.
Do not exceed		260°C

Wave Soldering Parameters	260°C Peak Temperature, 10 seconds max.



451/453 Series

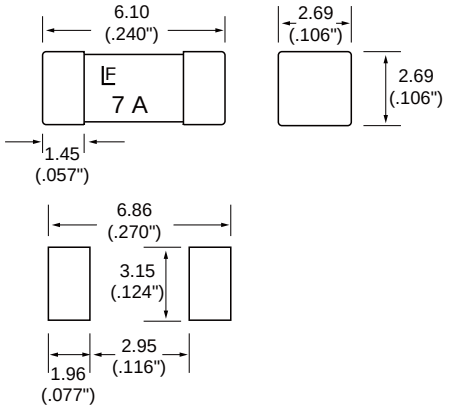
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Specifications are subject to change without notice.
Please refer to www.littelfuse.com/series/451.html or /453.html for current information.

Product Characteristics

Materials	**Body:** Ceramic **Terminations**: Gold-Plated Caps (for 451 RoHS/HF series) SnPb Plated Caps (for 451 Non-RoHS series) Silver-plated Caps (for 451 RoHS below 200mA Rating & 453 Series)
Product Marking	Brand, Ampere Rating
Operating Temperature	-55°C to 125°C
Moisture Sensitivity Level	Level 1, J-STD-020C
Solderability	MIL-STD-202, Method 208
Insulation Resistance (after Opening)	MIL-STD-202, Method 302, Test Condition A (10,000 ohms minimum)

Thermal Shock	MIL-STD-202, Method 107, Test Condition B, 5 cycles, -65°C / +125°C, 15 minutes @ each extreme
Mechanical Shock	MIL-STD-202, Method 213, Test I: Deenergized. 100G's pk amplitude, sawtooth wave 6ms duration, 3 cycles XYZ+xyz = 18 shocks
Vibration	MIL-STD-202, Method 201: 0.03" amplitude, 10-55 Hz in 1 min. 2hrs each XYZ=6hrs
Moisture Resistance	MIL-STD-202, Method 106, 10 cycles
Salt Spray	MIL-STD-202, Method 101, Test Condition B (48hrs)
Resistance to Soldering Heat	MIL-STD-202, Method 210, Test condition B (10 sec at 260°C)

Dimensions



Recommended pad layout

Part Numbering System

0451 001. M R L

SERIES
451 = Gold / SnPb / Silver Plated Caps
453 = Silver Plated Caps

AMP Code
Refer to Electrical characteristics table

QUANTITY Code
M = 1000 pcs
N = 5000 pcs

PACKAGING Code
R = Tape and Reel

RoHS Compliant & Halogen Free

NOTE: "L" suffix applies to 451 series only

- 451 series may be ordered as either "RoHS and HF" ("L" suffix) or non-RoHS (no suffix) version.
- 453 series is available only as RoHS compliant version and does not require "L" suffix. Please do not include "L" suffix within 453 series ordering instructions.
  453 series is only available from 200mA up to the highest rating specified. For ratings below 200mA, please use 451 series for ordering.

Packaging

Packaging Option	Packaging Specification	Quantity	Quantity & Packaging Code
12mm Tape and Reel	EIA RS-481-2 (IEC 286, part 3)	5000	NR
12mm Tape and Reel	EIA RS-481-2 (IEC 286, part 3)	1000	MR

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Specifications are subject to change without notice.
Please refer to www.littelfuse.com/series/451.html or /453.html for current information.

So the resistance value are marked as the following examples



1.96KΩ=196×10[1]Ω=29B



12.4Ω=124×10[-1]Ω=10X

(7) Standard E-24 and not belong to E-96 series values (≤±1%) of 0603 size: the marking is the same as 5% tolerance but marking as underli

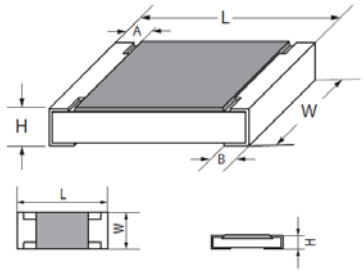


333=33KΩ



680=68Ω

5. Dimension



Type	Dimension(mm)				
	L	W	H	A	B
01005	0.40±0.02	0.20±0.02	0.13±0.02	0.10±0.05	0.10±0.03
0201	0.60±0.03	0.30±0.03	0.23±0.03	0.10±0.05	0.15±0.05
0402	1.00±0.10	0.50±0.05	0.35±0.05	0.20±0.10	0.25±0.10
0603	1.60±0.10	0.80±0.10	0.45±0.10	0.30±0.20	0.30±0.20
0805	2.00±0.15	1.25+0.15/-0.10	0.55±0.10	0.40±0.20	0.40±0.20
1206	3.10±0.15	1.55 +0.15/-0.10	0.55±0.10	0.45±0.20	0.45±0.20
1210	3.10±0.10	2.60±0.20	0.55±0.10	0.50±0.25	0.50±0.20
1812	4.50±0.20	3.20±0.20	0.55±0.20	0.50±0.20	0.50±0.20
2010	5.00±0.10	2.50±0.20	0.55±0.10	0.60±0.25	0.50±0.20
2512	6.35±0.10	3.20±0.20	0.55±0.10	0.60±0.25	0.50±0.20

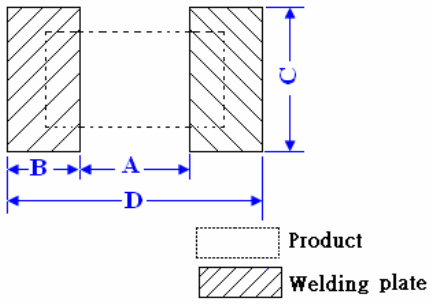
6. Resistance Range

Type	Power Rating	Resistance Range			
	at 70℃	0.5%	1.0%	2.0%	5.0%
01005	1/32W	---	10Ω-10MΩ	10Ω-10MΩ	1Ω -10MΩ
0201	1/20W	---	1Ω-10MΩ	1Ω-10MΩ	1Ω-10MΩ
0402	1/16W	1Ω-10MΩ	1Ω-10MΩ	1Ω-10MΩ	1Ω-10MΩ
0603	1/10W	1Ω-10MΩ	0.01Ω-10MΩ	0.01Ω-10MΩ	0.01Ω-10MΩ
0805	1/8W	1Ω-10MΩ	0.1Ω≤R＜10MΩ	0.1Ω≤R＜10MΩ	0.1Ω≤R＜10MΩ
	1/4W	---	0.01Ω≤R＜0.1Ω	0.01Ω≤R＜0.1Ω	0.01Ω≤R＜0.1Ω
1206	1/4W	1Ω-10MΩ	0.1Ω≤R＜10MΩ	0.1Ω≤R＜10MΩ	0.1Ω≤R＜10MΩ
	1/3W	---	0.01Ω≤R＜0.1Ω	0.01Ω≤R＜0.1Ω	0.01Ω≤R＜0.1Ω
1210	1/2W	1Ω-10MΩ	0.01Ω-10MΩ	0.01Ω-10MΩ	0.01Ω-10MΩ
1812	3/4W	1Ω-10MΩ	0.01Ω-10MΩ	0.01Ω-10MΩ	0.01Ω-10MΩ
2010	3/4W	1Ω-10MΩ	0.01Ω-10MΩ	0.01Ω-10MΩ	0.01Ω-10MΩ
2512	1W	1Ω-10MΩ	0.01Ω-10MΩ	0.01Ω-10MΩ	0.01Ω-10MΩ

7. Ratings

Type	Max. Working Voltage	Max. Overload Voltage	Dielectric withstanding Voltage	Resistance Value of Jumper	Rated Current of Jumper	Max. Overload Current of Jumper	Operating Temperature
01005	15V	30V	--	<50mΩ	0.5A	1A	-55℃~125℃
0201	25V	50V	--	<50mΩ	0.5A	1A	-55℃~155℃
0402	50V	100V	100V	<50mΩ	1A	2A	-55℃~155℃
0603	75V	150V	300V	<50mΩ	1A	2A	-55℃~155℃
0805	150V	300V	500V	<50mΩ	2A	5A	-55℃~155℃
1206	200V	400V	500V	<50mΩ	2A	10A	-55℃~155℃
1210	200V	500V	500V	<50mΩ	2A	10A	-55℃~155℃
1812	200V	500V	500V	<50mΩ	2A	10A	-55℃~155℃
2010	200V	500V	500V	<50mΩ	2A	10A	-55℃~155℃
2512	200V	500V	500V	<50mΩ	2A	10A	-55℃~155℃

8. Recommend the size of welding plate

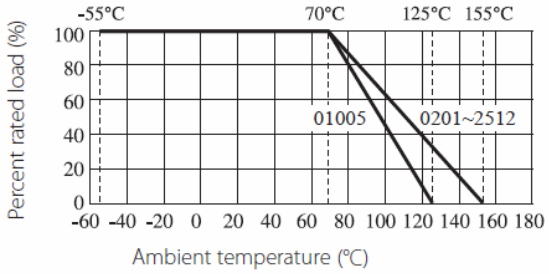


Type	Dimension(mm)			
	A	B	C	D
01005	0.14±0.03	0.2±0.03	0.2±0.03	0.54±0.03
0201	0.25±0.05	0.35±0.05	0.4±0.05	1.0±0.05
0402	0.50±0.05	0.45±0.05	0.5±0.05	1.4±0.05
0603	0.9±0.05	0.65±0.05	0.8±0.05	2.1±0.05
0805	1.0±0.1	1.0±0.1	1.3±0.1	3.0±0.1
1206	2.0±0.1	1.1±0.1	1.6±0.1	4.2±0.1
1210	2.0±0.1	1.1±0.1	2.6±0.1	4.2±0.1
1812	3.2±0.1	1.4±0.1	3.3±0.1	5.8±0.1
2010	3.6±0.1	1.3±0.1	2.6±0.1	6.2±0.1
2512	5.0±0.1	1.6±0.1	3.3±0.1	8.2±0.1

9. Derating Curve

Resistors shall have a power rating based on continuous load operation at an ambient temperature from -55℃ to 70℃. For temperature in excess of 70℃, the load shall be derated as shown in figure 1

Figure 1



Voltage rating:

Resistors shall have a rated direct-current (DC) continuous working

Voltage or an approximate sine-wave root-mean-square (RMS) alternating-current (AC) continuous working voltage at commercial-line frequency and waveform corresponding to the power rating, as determined from the following formula:

$RCWV = \sqrt{P \times R}$

Where: RCWV commercial-line frequency and waveform (Volt.)

P = power rating (WATT.) R = nominal resistance (OHM)

In no case shall the rated DC or RMS AC continuous working voltage be greater than the applicable maximum value.

The overload voltage is 2.5 times RCWV or Max. Overload voltage whichever is less

CONSTRUCTION

The resistor is constructed on top of a high-grade ceramic body. Internal metal electrodes are added on each end to make the contacts to the thick film resistive element. The composition of the resistive element is a noble metal imbedded into a glass and covered by a second glass to prevent environmental influences. The resistor is laser trimmed to the rated resistance value. The resistor is covered with a protective epoxy coat, finally the two external terminations (matte tin on Ni-barrier) are added, as shown in Fig.9.

Outlines

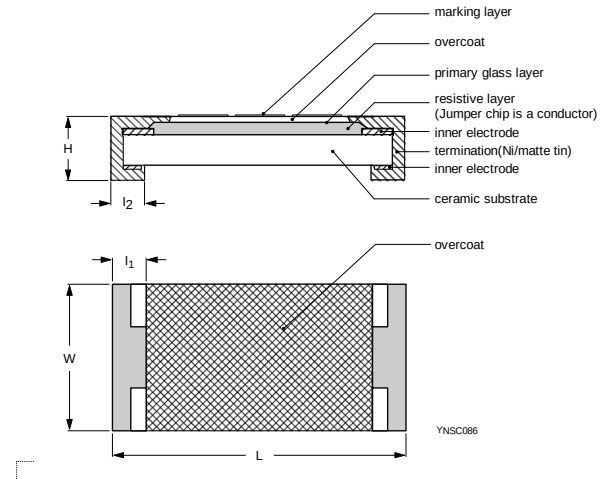


Fig. 9 Chip resistor outlines

DIMENSION

Table 1

TYPE	L (mm)	W (mm)	H (mm)	I_1 (mm)	I_2 (mm)
RC0075	0.30±0.01	0.15±0.01	0.13±0.01	0.08±0.03	0.08±0.03
RC0100	0.40±0.02	0.20±0.02	0.13±0.02	0.10±0.03	0.10±0.03
RC0201	0.60±0.03	0.30±0.03	0.23±0.03	0.10±0.05	0.15±0.05
RC0402	1.00±0.05	0.50±0.05	0.35±0.05	0.20±0.10	0.25±0.10
RC0603	1.60±0.10	0.80±0.10	0.45±0.10	0.25±0.15	0.25±0.15
RC0805	2.00±0.10	1.25±0.10	0.50±0.10	0.35±0.20	0.35±0.20
RC1206	3.10±0.10	1.60±0.10	0.55±0.10	0.45±0.20	0.45±0.20
RC1210	3.10±0.10	2.60±0.15	0.55±0.10	0.45±0.15	0.50±0.20
RC1218	3.10±0.10	4.60±0.10	0.55±0.10	0.45±0.20	0.40±0.20
RC2010	5.00±0.10	2.50±0.15	0.55±0.10	0.60±0.20	0.55±0.20
RC2512	6.35±0.10	3.10±0.15	0.55±0.10	0.60±0.20	0.60±0.20

ELECTRICAL CHARACTERISTICS

Table 2

CHARAC-TERISTICS	POWER	OPERATING TEMPERATURE RANGE	MAXIMUM WORKING VOLTAGE	MAXIMUM OVERLOAD VOLTAGE	DIELECTRIC WITHSTANDING VOLTAGE	RESISTANCE RANGE	TEMPERATURE COEFFICIENT	JUMPER CRITERIA
RC0075	1/50 W	-55℃ to 125℃	10V	25V	25V	5% (E24) 10Ω≦R≦1MΩ 1% (E24/E96) 10Ω≦R≦1MΩ Jumper<50mΩ	10Ω≦R<100Ω -200~+600ppm°C 100Ω≦R≦1MΩ ±200ppm°C	Rated Current 0.5A Maximum Current 1.0A
RC0100	1/32 W	-55℃ to 125℃	15V	30V	30V	5% (E24) 1Ω≦R≦22MΩ 1% (E24/E96) 1Ω≦R≦10MΩ 0.5% (E24/E96) 33Ω≦R≦470KΩ Jumper<50mΩ	1Ω≦R<10Ω -200~+600ppm°C 10Ω≤ R < 100Ω: ±300ppm/°C 100Ω≤ R ≤ 10MΩ: ±200ppm/°C 10MΩ< R ≤ 22MΩ: ±250ppm/°C	Rated Current 0.5A Maximum Current 1.0A

MARKING

RT0100 / RT0201 / RT0402 / RESISTANCE VALUE IS NOT IN E-24 / E96 SERIES

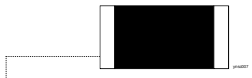


Fig. 1

No marking

RT0603



Fig. 2 Value = 56 kΩ

E-24 series: exception values 10/11/13/15/20/75 of E-24 series, one short bar under marking letter



Fig. 3 Value = 12.4 kΩ

E-96 series: including values 10/11/13/15/20/75 of E-24 series, 3 digits

RT0805 / RT1206 / RT1210 / RT2010 / RT2512



Fig. 4 Value = 10 kΩ

Either resistance in E-24 or E-96: 4 digits
First three digits for significant figure and 4th digit for number of zeros

For further marking information, please see special data sheet "Chip resistors marking".

CONSTRUCTION

The resistors are constructed out of a high-grade ceramic body. Internal metal electrodes are added at each end and connected by a resistive layer. The resistive layer is adjusted to give the approximate required resistance and laser cutting of this resistive layer that achieves tolerance trims the value. The resistive layer is covered with a protective coat and printed with the resistance value. Finally, the two external terminations (matte tin) are added. See fig. 5.

DIMENSION

Table 1 For outlines see fig. 5

TYPE	L (mm)	W (mm)	H (mm)	I_1 (mm)	I_2 (mm)
RT0100	0.40±0.02	0.20±0.02	0.13±0.02	0.10±0.03	0.10±0.03
RT0201	0.60 ±0.03	0.30 ±0.03	0.23 ±0.03	0.10 ±0.05	0.15 ±0.05
RT0402	1.00 ±0.10	0.50 ±0.05	0.30 ±0.05	0.20 ±0.10	0.25 ±0.10
RT0603	1.60 ±0.10	0.80 ±0.10	0.45 ±0.10	0.25 ±0.15	0.25 ±0.15
RT0805	2.00 ±0.10	1.25 ±0.10	0.50 ±0.10	0.35 ±0.20	0.35 ±0.20
RT1206	3.10 ±0.10	1.60 ±0.10	0.55 ±0.10	0.45 ±0.20	0.40 ±0.20
RT1210	3.10 ±0.10	2.60 ±0.15	0.55 ±0.10	0.50 ±0.20	0.50 ±0.20
RT2010	5.00 ±0.10	2.50 ±0.15	0.55 ±0.10	0.60 ±0.20	0.50 ±0.20
RT2512	6.35 ±0.10	3.20 ±0.15	0.55 ±0.10	0.60 ±0.20	0.50 ±0.20

OUTLINES

For dimension see Table 1

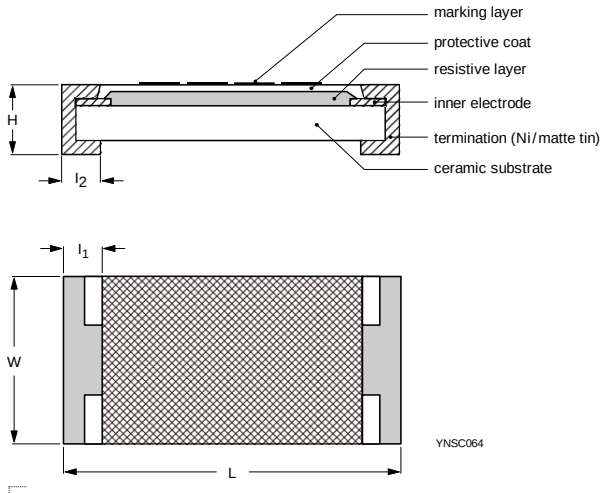


Fig. 5 Chip resistor outlines

CONSTRUCTION

The capacitor consists of a rectangular block of ceramic dielectric in which a number of interleaved metal electrodes are contained. This structure gives rise to a high capacitance per unit volume.

The inner electrodes are connected to the two end terminations and finally covered with a layer of plated tin (NiSn). The terminations are lead-free. A cross section of the structure is shown in Fig.1.

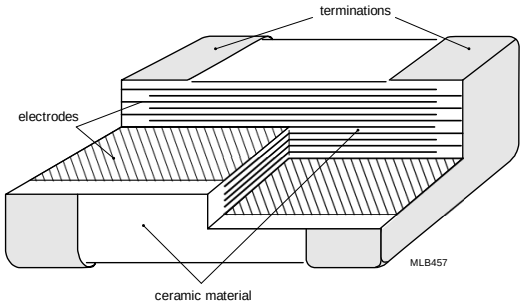


Fig. 1 Surface mounted multilayer ceramic capacitor construction

DIMENSION

Table 1 For outlines see fig. 2

TYPE	L_1 (mm)	W (mm)	T (MM)	L_2 / L_3 (mm) min.	L_2 / L_3 (mm) max.	L_4 (mm) min.
0201	0.6 ±0.03	0.3±0.03	0.3±0.03	0.10	0.20	0.20
0402	1.0 ±0.05	0.5 ±0.05	0.5 ±0.05	0.15	0.35	0.30
0603	1.6 ±0.10	0.8 ±0.10	0.8 ±0.10	0.20	0.60	0.40
0805	2.0 ±0.10	1.25 ±0.10	0.6 ±0.10	0.25	0.75	0.70
	2.0 ±0.20	1.25 ±0.20	0.85 ±0.10			
			1.25 ±0.20			
1206	3.2 ±0.15	1.6 ±0.15	0.6 ±0.10	0.25	0.75	1.40
			0.85 ±0.10			
	3.2 ±0.30	1.6 ±0.20	0.85 ± 0.10[1]			
			1.6 ±0.20			
	3.2 ±0.30	1.6 ±0.30	1.6 ±0.30	0.3	0.9	1.4
1210	3.2 ±0.20	2.5 ±0.20	0.85 ±0.10	0.25	0.75	1.40
	3.2 ±0.40	2.5 ±0.30	1.25 ±0.20			
			1.6 ±0.20			
			2.0 ±0.20			
1808	4.5 ±0.40	2.0 ±0.30	1.25 ±0.20	0.25	0.75	2.20
1812	4.5 ±0.40	3.2 ±0.30	0.85 ±0.10	0.25	0.75	2.20
			1.25 ±0.20			
			1.60 ±0.20			
2020	5.7 ±0.40	5.0 ±0.30	2.0 ±0.20	0.25	0.75	3.40

NOTE

1. Dimension for size 1206, 12nF to 33nF, 25V to 100V.

OUTLINES

For dimension see Table 1

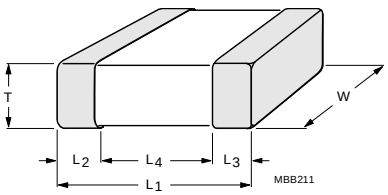


Fig. 2 Surface mounted multilayer ceramic c

CAPACITANCE RANGE & THICKNESS FOR X7R

Table 4 Sizes from 0805

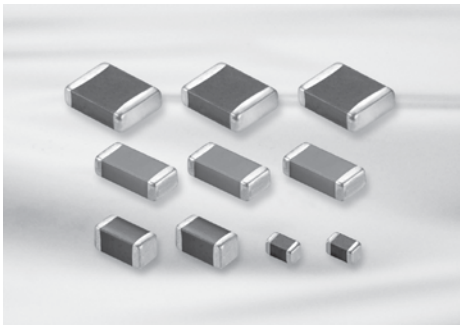
CAP.	0805							
	6.3 V	10 V	16 V	25 V	50 V	100V	200V	250V
100 pF	E0	E0	E0	E0	E0	E0	EA	EA
150 pF	E0	E0	E0	E0	E0	E0	EA	EA
220 pF	E0	E0	E0	E0	E0	E0	EA	EA
330 pF	E0	E0	E0	E0	E0	E0	EA	EA
470 pF	E0	E0	E0	E0	E0	E0	EA	EA
680 pF	E0	E0	E0	E0	E0	E0	EA	EA
1.0 nF	E0	E0	E0	E0	E0	E0	EA	EA
1.5 nF	E0	E0	E0	E0	E0	E0	EA	EA
2.2 nF	E0	E0	E0	E0	E0	E0	EA	EA
3.3 nF	E0	E0	E0	E0	E0	E0	EA	EA
4.7 nF	E0	E0	E0	E0	E0	E0	EA	EA
6.8 nF	E0	E0	E0	E0	E0	E0	EB	EB
10 nF	E0	E0	E0	E0	E0	E0	EB	EB
15 nF	E0	E0	E0	E0	E0	EA	EB	EB
22 nF	E0	E0	E0	E0	E0	EA	EB	EB
33 nF	EA	EA	EA	EA	EA	EB	EB	EB
47 nF	EA	EA	EA	EA	EA	EB	EB	EB
68 nF	EA	EA	EA	EA	EA	EB	EB	EB
100 nF	EA	EA	EA	EA	EA	EB	EB	EB
150 nF	EA	EA	EA	EA	EA	EB		
220 nF	EA	EA	EA	EA	EB	EB		
330 nF	EB	EB	EB	EB	EB	EB		
470 nF	EB	EB	EB	EB	EB	EB		
680 nF	EB	EB	EB	EB	EB	EB		
1 µF	EB	EB	EB	EB	EB	EB		
2.2 µF	EB	EB	EB	EB	EB			
4.7 µF	EB	EB	EB	EB				
10 µF	EB	EB	EB					

NOTE

1. Values in shaded cells indicate thickness class in mm
2. Capacitance value of non E-6 series is on request
3. For product with 5% tolerance, please contact local sales force before ordering

CASE SIZE	L (mm)	W (mm)	T (mm)	DIMENSION CODE
0805	2.0 ±0.1	1.25 ±0.1	0.6 ±0.1	E0
	2.0 ±0.2	1.25 ±0.2	0.85 ±0.1	EA
	2.0 ±0.2	1.25 ±0.2	1.25 ±0.2	EB

Standard & High Capacitors



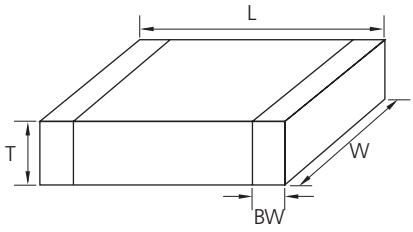
Feature

- Wide selection of size : from 0402 to 2220
- Highly reliable tolerance and high speed automatic chip placement on PCBs
- Wide capacitance range
- Wide temperature compensation and voltage range
  : from C0G to Y5V and from 6.3V to 50V
- Highly reliable performance
- Highly resistant termination metal
- Tape & reel for surface mount assembly

Application

- HHP, DSC, DVC, LCD, TV, Memory Module, PDA, Game Machine
- Desktop PC, Note PC, HHP, DC-DC Converter, DSC
- Tuner (Product code C is suitable.)

Structure and Dimensions



Size Code	EIA Code	Dimension(㎜)				
		L	W	T	Thickness Code	BW
05	0402	1.00±0.05	0.50±0.05	0.50±0.05	5	0.25±0.10
10	0603	1.60±0.10	0.80±0.10	0.50+0.0/–0.1(*)	5	0.30±0.20
		1.60±0.10	0.80±0.10	0.80±0.10	8	
21	0805	2.00±0.10	1.25±0.10	0.85±0.10	C	0.5+0.2/–0.3
		2.00±0.10	1.25±0.10	1.15±0.10	M	
		2.00±0.10	1.25±0.10	1.25±0.10	F	
		2.00±0.15	1.25±0.15	1.25±0.15	Q	
		2.00±0.20	1.25±0.20	1.25±0.20	Y	
31	1206	3.20±0.20	1.60±0.20	0.60±0.10(*)	6	0.50±0.30
		3.20±0.15	1.60±0.15	0.85±0.15	C	
		3.20±0.20	1.60±0.20	0.85±0.10(*)	C	
		3.20±0.20	1.60±0.20	1.15±0.10(*)	P	
		3.20±0.15	1.60±0.15	1.25±0.15	F	
		3.20±0.20	1.60±0.20	1.60±0.20	H	
32	1210	3.20±0.30	2.50±0.20	0.85±0.10(*)	C	0.60±0.30
		3.20±0.30	2.50±0.20	0.90±0.10(*)	9	
		3.20±0.30	2.50±0.20	1.60±0.20	H	
		3.20±0.30	2.50±0.20	1.80±0.20(*)	U	
		3.20±0.30	2.50±0.20	2.00±0.20	I	
		3.20±0.30	2.50±0.20	2.50±0.20	J	
		3.20±0.40	2.50±0.30	2.50±0.30	V	
42	1808	4.50±0.40	2.00±0.20	1.25±0.20	F	0.80±0.30
		4.50±0.40	2.00±0.20	1.40±0.20	G	
		4.50±0.40	2.00±0.20	2.00±0.20	I	
43	1812	4.50±0.40	3.20±0.30	1.25±0.20	F	0.80±0.30
		4.50±0.40	3.20±0.30	2.50±0.20	J	
		4.50±0.40	3.20±0.30	3.20±0.30	L	
55	2220	5.70±0.40	5.00±0.40	2.50±0.20	J	1.00±0.30
		5.70±0.40	5.00±0.40	3.20±0.30	L	

■ * Mark is only applicable to "L"code, 12th code in part number.

SAMSUNG ELECTRO-MECHANICS

RoHS+Halogen Compliant ecoparts

SPECIFICATION (Reference sheet)

· **Supplier : Samsung electro-mechanics**
· **Product : Multi-layer Ceramic Capacitor**

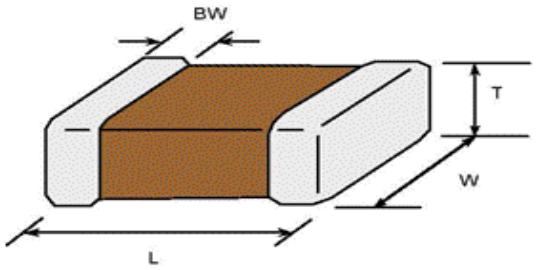
· **Samsung P/N :** CL32B106KBJNNWE
· **Description :** CAP, 10uF, 50V, ±10%, X7R, 1210

A. Samsung Part Number

CL	32	B	106	K	B	J	N	N	W	E
①	②	③	④	⑤	⑥	⑦	⑧	⑨	⑩	⑪

①	**Series**	Samsung Multi-layer Ceramic Capacitor			
②	**Size**	1210 (inch code)	L: 3.20 ± 0.30 ㎜	W:	2.50 ± 0.20 ㎜
③	**Dielectric**	X7R	⑧ **Inner electrode**		Ni
④	**Capacitance**	10 uF	**Termination**		Cu
⑤	**Capacitance tolerance**	±10 %	**Plating**		Sn 100% (Pb Free)
			⑨ **Product**		Normal
⑥	**Rated Voltage**	50 V	⑩ **Special**		Industrial (Network,etc)
⑦	**Thickness**	2.50 ± 0.20 ㎜	⑪ **Packaging**		Embossed Type, 7" reel

B. Structure & Dimension



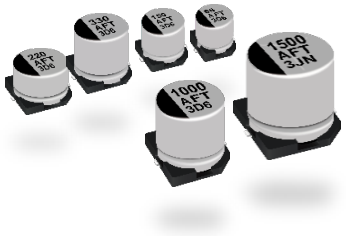
Samsung P/N	Dimension(㎜)			
	L	W	T	BW
CL32B106KBJNNWE	3.20 ± 0.30	2.50 ± 0.20	2.50 ± 0.20	0.60 ± 0.30

Panasonic
INDUSTRY

Aluminum Electrolytic Capacitors

Surface Mount Type

FT series **High temperature Lead-Free reflow**



Features

- Endurance : 105 °C 2000 h to 5000 h
- Miniaturized, Low ESR (1 size smaller than series FK)
- Vibration-proof product (30G guaranteed) is available upon request (ø6.3 ≤)
- AEC-Q200 compliant
- RoHS compliant

Specifications

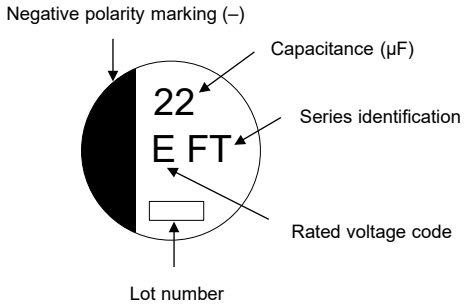
Category temp. range	–55 °C to +105 °C	
Rated voltage range	6.3 V to 50 V	
Capacitance range	10 µF to 2200 µF	
Capacitance tolerance	±20 % (120 Hz / +20 °C)	
Leakage current	I ≤ 0.01 CV（µA） After 2 minutes	
Dissipation factor (tan δ)	Please see the attached characteristics list	
Endurance	After applying rated working voltage for 2000 hours at +105 °C ± 2 °C and then being stabilized at +20 °C, capacitors shall meet the following limits. (Suffix “G” in 6.3 V : 3000 hours, 10 V to 50 V : 5000 hours)	
	Capacitance change	Within ±30 % of the initial value (Suffix “G” is ≤ ±35 %)
	Dissipation factor (tan δ)	≤ 200 % of the initial limit (Suffix “G” is ≤ 300 %)
	Leakage current	Within the initial limit
Shelf life	After storage for 1000 hours at +105 °C ± 2 °C with no voltage applied and then being stabilized at +20 °C, capacitors shall meet the following limits. (Capacitors are left at room temperature and humidity for 2 hours after the test. Then, apply the rated voltage for 30min with a series protection resistance of 1000 Ω ± 10 Ω. Measure electrical characteristics after discharging the capacitor.)	
	Capacitance change	Within ±30 % of the initial value
	Dissipation factor (tan δ)	≤ 200 % of the initial limit
	Leakage current	Within the initial limit
Resistance to soldering heat	After reflow soldering and then being stabilized at +20 °C, capacitors shall meet the following limits.	
	Capacitance change	Within ±10 % of the initial value
	Dissipation factor (tan δ)	Within the initial limit
	Leakage current	Within the initial limit

Frequency correction factor for ripple current

Cap. (µF) \ Freq. (Hz)	120	1 k	10 k	100 k to
10 to 470	0.65	0.85	0.95	1.00
560 to 2200	0.70	0.90	0.95	1.00

Marking

Example : 25 V 22 µF
Marking color : BLACK

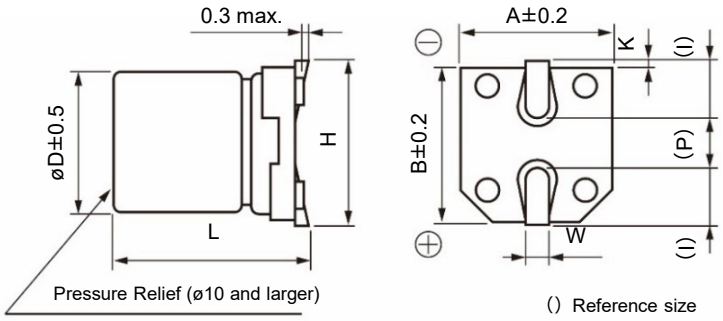


R.voltage code

Unit : V

j	6.3	E	25
A	10	V	35
C	16	H	50

Dimensions



Unit : mm

Size code	øD	L	A、B	H	I	W	P	K
B	4.0	5.8±0.3	4.3	5.5 max.	1.8	0.65±0.1	1.0	$0.35^{+0.15}_{-0.20}$
C	5.0	5.8±0.3	5.3	6.5 max.	2.2	0.65±0.1	1.5	$0.35^{+0.15}_{-0.20}$
D	6.3	5.8±0.3	6.6	7.8 max.	2.6	0.65±0.1	1.8	$0.35^{+0.15}_{-0.20}$
D8	6.3	7.7±0.3	6.6	7.8 max.	2.6	0.65±0.1	1.8	$0.35^{+0.15}_{-0.20}$
F	8.0	10.2±0.3	8.3	10.0 max.	3.4	0.90±0.2	3.1	0.70±0.2
G	10.0	10.2±0.3	10.3	12.0 max.	3.5	0.90±0.2	4.6	0.70±0.2

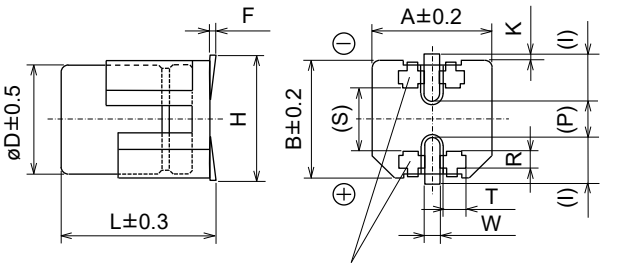
・The dimensions of the vibration-proof products, please refer to the page of the mounting specification.

Design and specifications are each subject to change without notice. Ask factory for the current technical specifications before purchase and/or use.
Should a safety concern arise regarding this product, please be sure to contact us immediately.

Dimensions (Vibration-proof products)

✻ The size and shape are different from standard products.

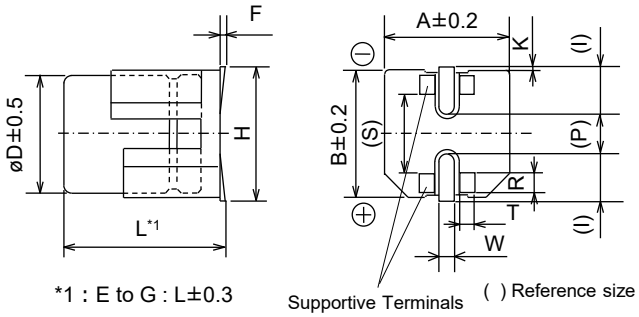
< Size code : D, D8 >



Supportive Terminals ( ) Reference size

※The diagram is not drawn to scale.

< Size code : E, F, G, H13, J16, K16, K21 >



*1 : E to G : L±0.3
H13 to K21 : L±0.5

Supportive Terminals ( ) Reference size

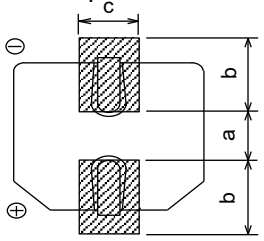
Unit : mm

Size code	øD	L	A, B	H max.	F	I	W	P	K	R	S	T
D	6.3	6.1	6.6	7.8	0 to +0.15	2.4	0.65±0.1	2.2	0.35 $^{+0.15}_{-0.20}$	1.1±0.2	3.3	1.05±0.2
D8	6.3	8.0	6.6	7.8	0 to +0.15	2.4	0.65±0.1	2.2	0.35 $^{+0.15}_{-0.20}$	1.1±0.2	3.3	1.05±0.2
E	8.0	6.5	8.3	9.5	0 to +0.15	3.4	0.7±0.1	2.2	0.35 $^{+0.15}_{-0.20}$	0.70±0.2	5.3	1.7±0.2
F	8.0	10.5	8.3	10.0	0 to +0.15	3.4	1.2±0.2	3.1	0.70±0.2	0.70±0.2	5.3	1.3±0.2
G	10.0	10.5	10.3	12.0	0 to +0.15	3.5	1.2±0.2	4.6	0.70±0.2	0.70±0.2	6.9	1.3±0.2
H13	12.5	13.8	13.5	15.0	–0.1 to +0.15	4.7	1.2±0.2	4.4	0.70±0.3	2.2±0.2	7.1	2.4±0.2
J16	16.0	16.8	17.0	19.0	–0.1 to +0.15	5.5	1.4±0.2	6.7	0.70±0.3	3.0±0.2	9.0	1.9±0.2
K16	18.0	16.8	19.0	21.0	–0.1 to +0.15	6.7	1.4±0.2	6.7	0.70±0.3	3.0±0.2	11.0	1.9±0.2
K21	18.0	21.8	19.0	21.0	–0.1 to +0.15	6.7	1.4±0.2	6.7	0.70±0.3	3.0±0.2	11.0	1.9±0.2

Land / Pad pattern

The circuit board land/pad pattern size for chip capacitors is specified in the following table.
The land pitch influences installation strength.

● Standard products



Land space



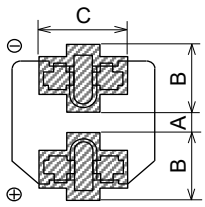
(Table of board land size vs. capacitor size) Unit : mm

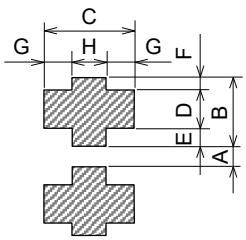
Size code	a	b	c
B (ø4)	1.0	2.5	1.6
C (ø5)	1.5	2.8	1.6
D (ø6.3)	1.8	3.2	1.6
D8 (ø6.3x7.7L)	1.8	3.2	1.6
E (ø8x6.2L)	2.2	4.0	1.6
F (ø8x10.2L)	3.1	4.0	2.0
G (ø10x10.2L)	4.6	4.1	2.0
H13 (ø12.5)	3.9	6.2	2.0
J16 (ø16)	6.0	6.9	2.5
K16, K21 (ø18)	6.0	7.9	2.5

Larger dimension of "a" may prevent back fillet from being formed adequately to obtain required solder strength.

● Vibration-proof products

< Size code : D, D8 >



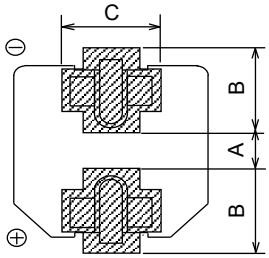


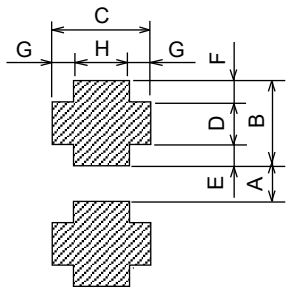
(Table of board land size vs. capacitor size) Unit : mm

Size code	A	B	C	D	E	F	G	H
D (ø6.3xL6.1)	1.2	3.6	3.2	2.0	1.0	0.7	1.0	1.2
D8 (ø6.3xL8.0)	1.2	3.6	3.2	2.0	1.0	0.7	1.0	1.2
E (ø8x6.5L)	1.8	4.2	5.0	1.3	1.5	1.4	1.5	2.0
F (ø8x10.5L)	2.7	4.0	4.7	1.3	1.0	1.7	1.1	2.5
G (ø10)	3.9	4.4	4.7	1.3	1.2	1.9	1.1	2.5
H13 (ø12.5)	3.9	6.2	6.9	2.8	1.3	2.1	2.2	2.5
J16 (ø16)	5.8	7.0	6.2	3.6	1.3	2.1	1.7	2.8
K16, K21 (ø18)	5.8	8.0	6.2	3.6	2.3	2.1	1.7	2.8

Larger dimension of "a" may prevent back fillet from being formed adequately to obtain required solder strength.

< Size code : E, F, G, H, J, K >





∗ Follow your design rules for mounting conditions, solderability, and fitting strength if some exists in order to determine the land pattern.

∗ The vibration-proof capacitors of size ø6.3 has support terminals extending from the bottom side to the lead edge. Then, make sure to find appropriate soldering conditions to form fillet on the support terminals if required for appearance inspection.

CERAMIC SMD CRYSTAL

ABM3B



RoHS Compliant

5.0 X 3.2 X 1.1mm

OPTIONS AND PART IDENTIFICATION: (Left blank if standard)

ABM3B □ – ▭ MHz - □ - ▭ - □ - □ - □ - □ - □

Height (mm)
Blank: 1.1mm max
1: 0.8mm max

Frequency in MHz
e.g. 8.000MHz
14.31818MHz
26.000MHz
125.000MHz

Load Capacitance
Please specify CL(6 to 32pF) in pF or S for Series
Please contact ABRACON for values under 6pF

Custom ESR if other than standard
R□
□: Specify a value in Ω (e.g.: R40)

Operating Temp.
E: 0°C ~ +70°C
B: -20°C ~ +70°C
C: -30°C ~ +70°C
N: -30°C ~ +85°C
D: -40°C ~ +85°C
J****: -40°C ~ +105°C
K****: -40°C ~ +125°C
L*: -55°C ~ +125°C

Freq. Tolerance
1: ± 10 ppm
7: ± 15 ppm
2: ± 20 ppm
3: ± 25 ppm
4: ± 30 ppm

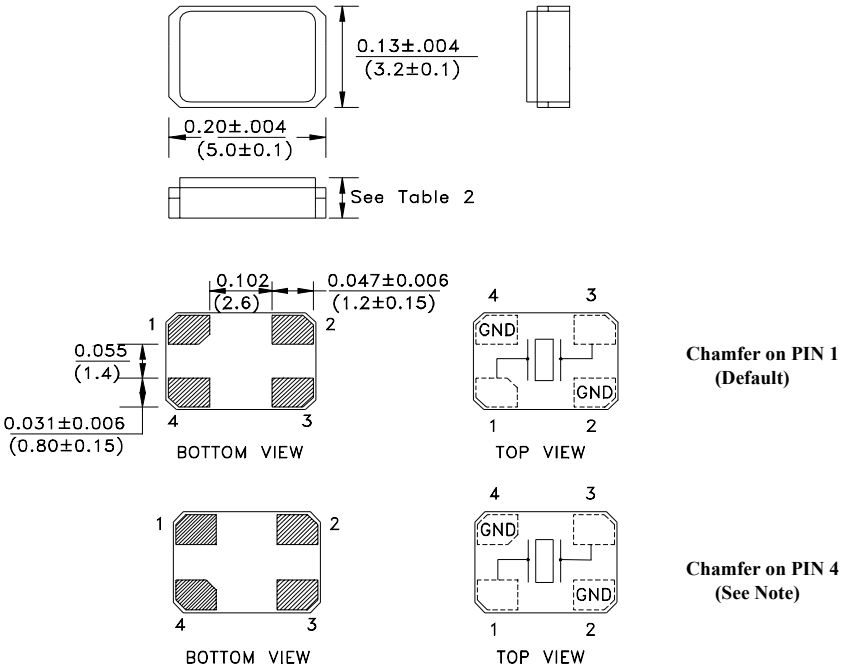
Freq. Stability
U**: ± 10 ppm
G***: ± 15 ppm
X: ± 20 ppm
W: ± 25 ppm
Y: ± 30 ppm
H: ± 35 ppm
Z: ± 100 ppm

(*) Option Z only
(**)Standard temp or options E and B only
(***)Standard temp or options E, B, C and N only.
(****) Freq. stability ±50ppm and ±100ppm only. Contact ABRACON for tighter freq. stability.

Packaging
Blank: Bulk
T: 1k pcs per reel
T2: 250 pcs per reel

OUTLINE DIMENSIONS:

Please see Table 2 below for height and chamfer information.



Chamfer on PIN 1 (Default)

Chamfer on PIN 4 (See Note)

Recommended land pattern

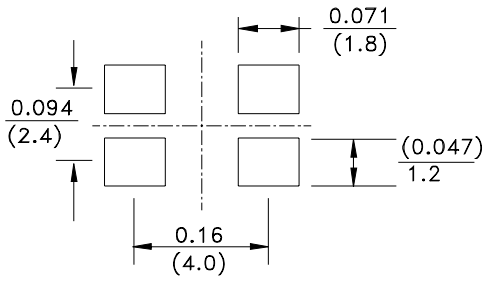


Table 2

Height (mm)		Chamfer PIN #
ABM3B	1.1mm max	#1 (Default), #4 (See Note)
ABM3B1	0.8mm max	#1 (Default), #4 (See Note)

Note: Due to the availability of raw materials, this part may be manufactured with the chamfer on pin 4. Be advised that this does not affect the electrical characteristics of the crystal in any way.

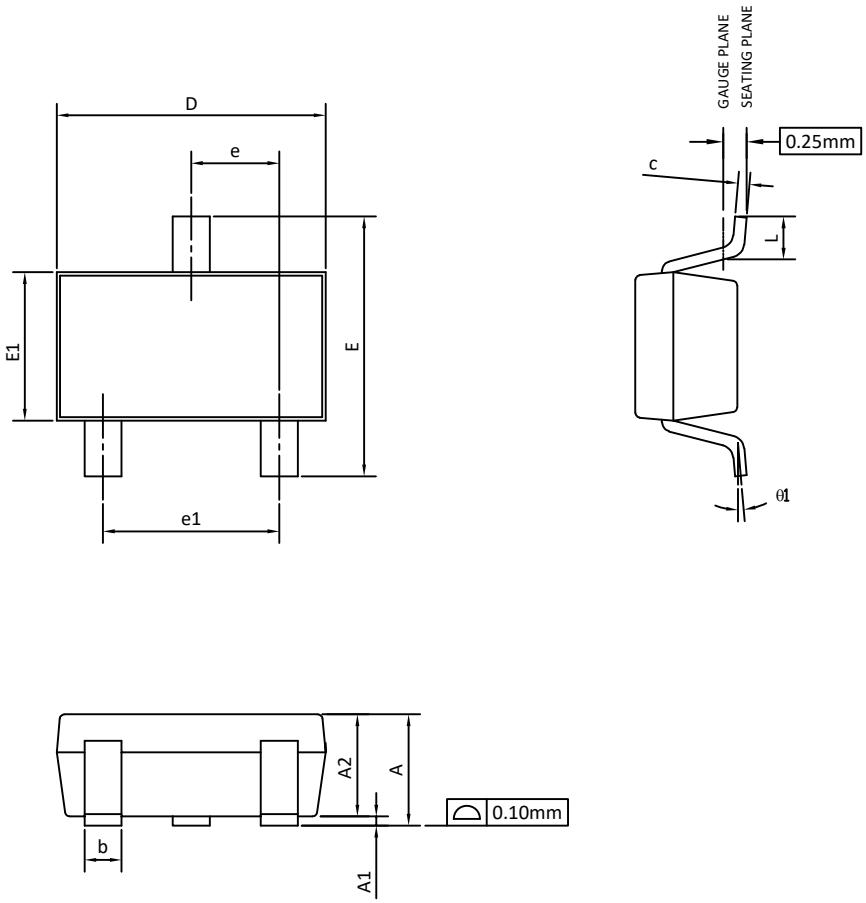
Dimension : Inches (mm)

2 Faraday, Suite# B | Irvine | CA 92618 **Revised: 08.25.15**
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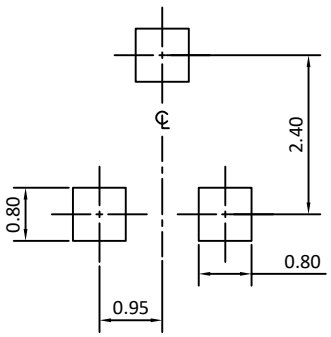
ALPHA & OMEGA SEMICONDUCTOR

Document No.	PO-00001
Version	N

SOT23 PACKAGE OUTLINE



RECOMMENDED LAND PATTERN



UNIT: mm

SYMBOLS	DIMENSION IN MM			DIMENSION IN INCHES		
	MIN	NOM	MAX	MIN	NOM	MAX
A	0.85	---	1.25	0.033	----	0.049
A1	0.00	---	0.15	0.000	----	0.006
A2	0.70	1.05	1.20	0.028	0.041	0.047
b	0.30	0.40	0.50	0.012	0.016	0.020
c	0.08	0.13	0.20	0.003	0.005	0.008
D	2.80	2.90	3.10	0.110	0.114	0.122
e	0.95 BSC			0.037 BSC		
e1	1.90 BSC			0.075 BSC		
E	2.60	2.80	3.00	0.102	0.110	0.118
E1	1.40	1.60	1.80	0.055	0.063	0.071
L	0.30	---	0.60	0.012	----	0.024
Q1	0°	5°	8°	0°	5°	8°

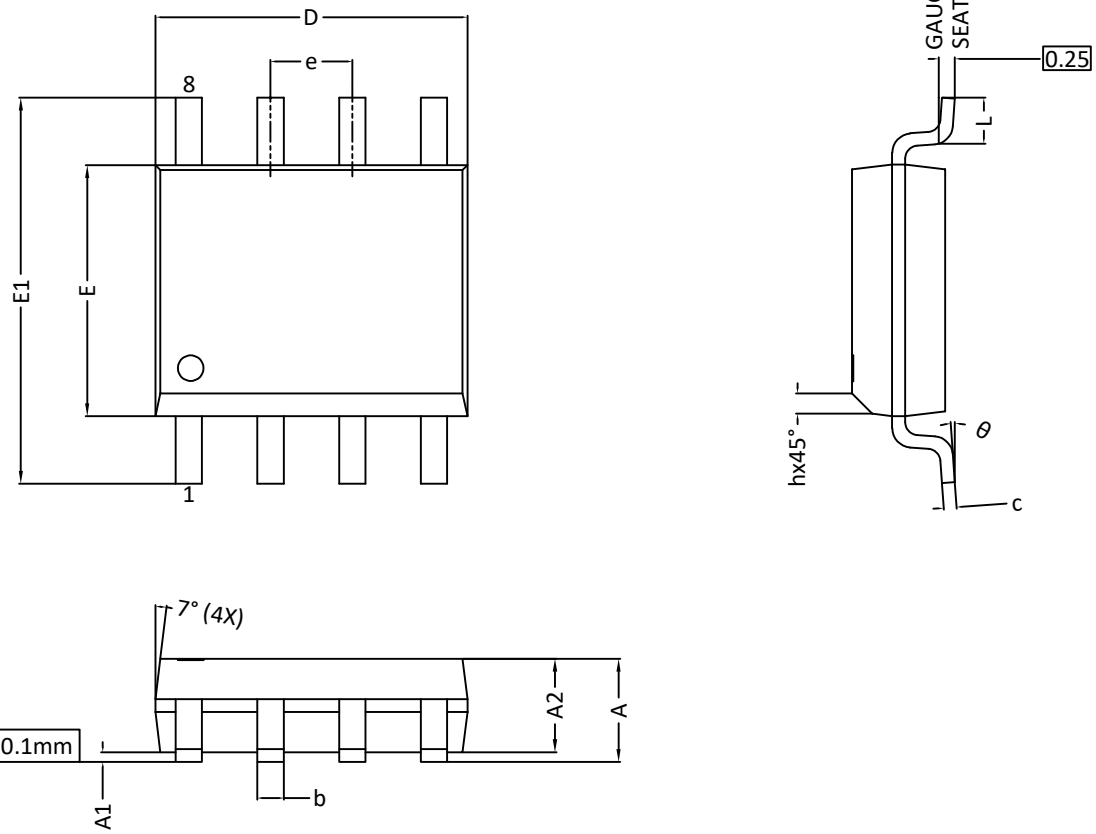
NOTE:

1. PACKAGE BODY SIZES EXCLUDE MOLD FLASH OR GATE BURRS.
   MOLD FLASH AT THE NON-LEAD SIDES SHOULD BE LESS THAN 5 MILS EACH.
2. TOLERANCE ±0.100 mm (4 mil) UNLESS OTHERWISE SPECIFIED.
3. DIMENSION L IS MEASURED IN GAUGE PLANE.
4. CONTROLLING DIMENSION IS MILLIMETER. CONVERTED INCH DIMENSIONS
   ARE NOT NECESSARILY EXACT.
5. ALL DIMENSIONS ARE IN MILLIMETERS.

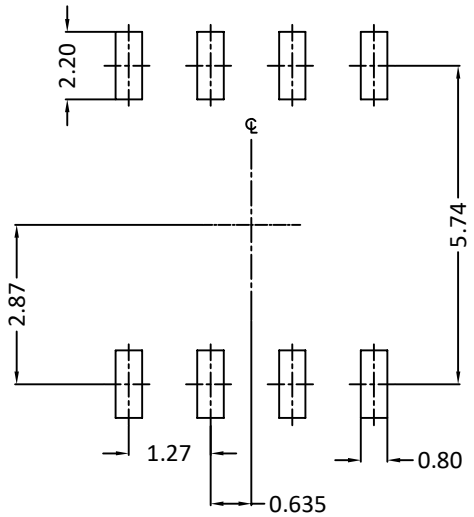
ALPHA & OMEGA SEMICONDUCTOR

Document No.	PO-00004
Version	L

SO8(SOP-8L) PACKAGE OUTLINE



RECOMMENDED LAND PATTERN



SYMBOLS	DIMENSION IN MM			DIMENSION IN INCHES		
	MIN	NOM	MAX	MIN	NOM	MAX
A	1.35	1.65	1.75	0.053	0.065	0.069
A1	0.10	0.15	0.25	0.004	0.006	0.010
A2	1.25	1.50	1.65	0.049	0.059	0.065
b	0.31	0.41	0.51	0.012	0.016	0.020
c	0.17	0.20	0.25	0.007	0.008	0.010
D	4.80	4.90	5.00	0.189	0.193	0.197
E	3.80	3.90	4.00	0.150	0.154	0.157
E1	5.80	6.00	6.20	0.228	0.236	0.244
e	1.27 BSC			0.050 BSC		
h	0.25	0.30	0.50	0.010	0.012	0.020
L	0.40	0.69	1.27	0.016	0.027	0.050
θ	0°	4°	8°	0°	4°	8°

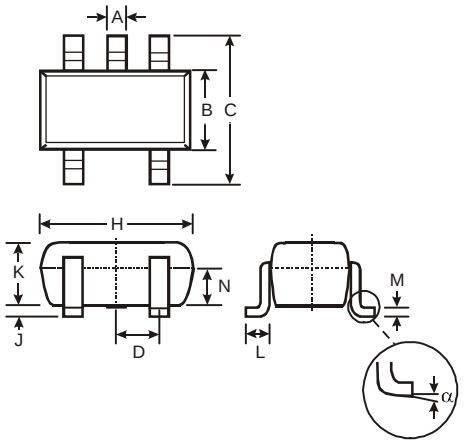
UNIT: mm

NOTE
1. ALL DIMENSIONS ARE IN MILLMETERS.
2. DIMENSIONS ARE INCLUSIVE OF PLATING.
3. PACKAGE BODY SIZES EXCLUDE MOLD FLASH AND GATE BURRS.
   MOLD FLASH AT THE NON-LEAD SIDES SHOULD BE LESS THAN 6 MILS EACH.
4. DIMENSION L IS MEASURED IN GAUGE PLANE.
5. CONTROLLING DIMENSION IS MILLIMETER.
   CONVERTED INCH DIMENSIONS ARE NOT NECESSARILY EXACT.

Package Outline Dimensions (Previously identified as SOT-23-5 for this product)

Please see http://www.diodes.com/package-outlines.html for the latest version.

SOT25

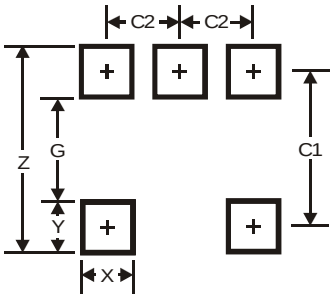


SOT25			
Dim	**Min**	**Max**	**Typ**
A	0.35	0.50	0.38
B	1.50	1.70	1.60
C	2.70	3.00	2.80
D	-	-	0.95
H	2.90	3.10	3.00
J	0.013	0.10	0.05
K	1.00	1.30	1.10
L	0.35	0.55	0.40
M	0.10	0.20	0.15
N	0.70	0.80	0.75
α	0°	8°	-
All Dimensions in mm			

Suggested Pad Layout

Please see http://www.diodes.com/package-outlines.html for the latest version.

SOT25



Dimensions	Value
Z	3.20
G	1.60
X	0.55
Y	0.80
C1	2.40
C2	0.95

Note: The suggested land pattern dimensions have been provided for reference only, as actual pad layouts may vary depending on application. These dimensions may be modified based on user equipment capability or fabrication criteria. A more robust pattern may be desired for wave soldering and is calculated by adding 0.2 mm to the 'Z' dimension. For further information, please reference document IPC-7351A, Naming Convention for Standard SMT Land Patterns, and for International grid details, please see document IEC, Publication 97.

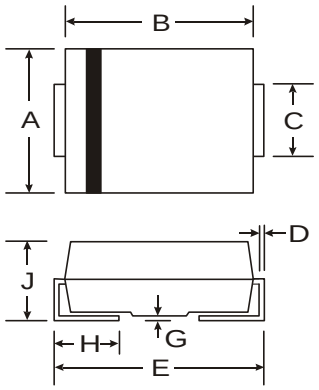
Note: For high voltage applications, the appropriate industry sector guidelines should be considered with regards to creepage and clearance distances between device Terminals and PCB tracking.

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Package Outline Dimensions

Please see AP02002 at http://www.diodes.com/datasheets/ap02002.pdf for the latest version.

SMA/SMB



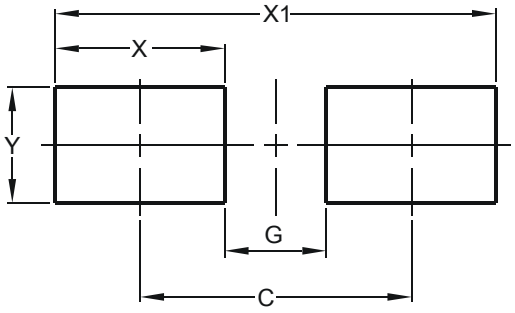
SMA		
Dim	**Min**	**Max**
A	2.29	2.92
B	4.00	4.60
C	1.27	1.63
D	0.15	0.31
E	4.80	5.59
G	0.05	0.20
H	0.76	1.52
J	2.01	2.30
All Dimensions in mm		

SMB		
Dim	**Min**	**Max**
A	3.30	3.94
B	4.06	4.57
C	1.96	2.21
D	0.15	0.31
E	5.00	5.59
G	0.05	0.20
H	0.76	1.52
J	2.00	2.50
All Dimensions in mm		

Suggested Pad Layout

Please see AP02001 at http://www.diodes.com/datasheets/ap02001.pdf for the latest version.

SMA/SMB

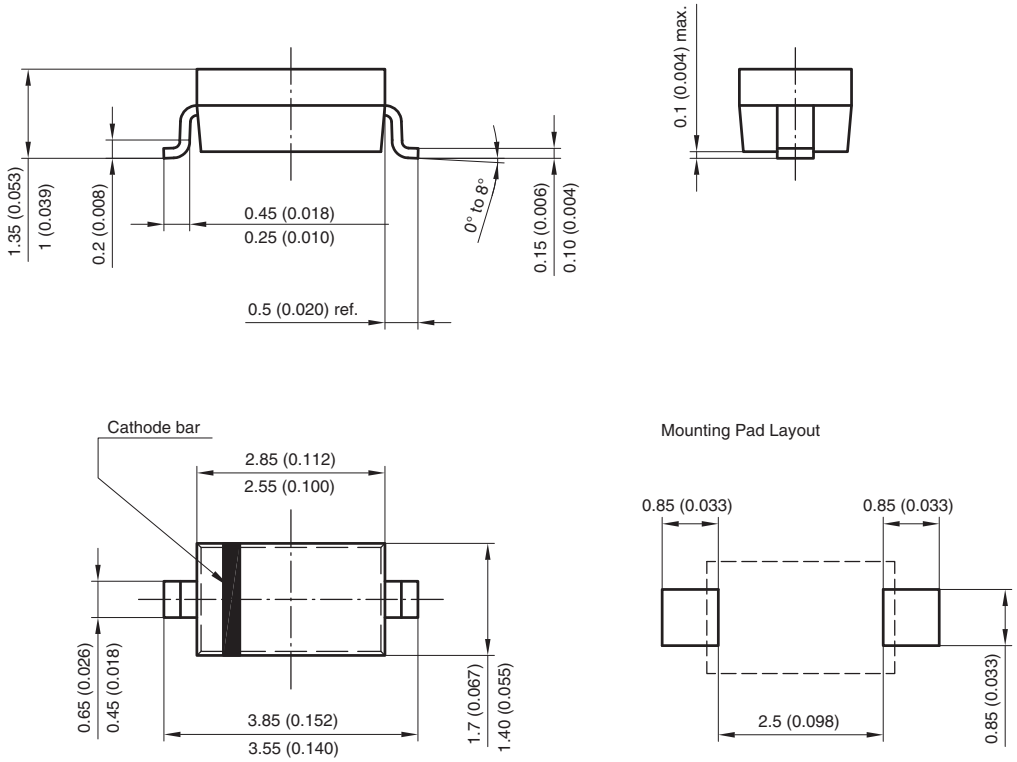


SMA	
Dimensions	**Value (in mm)**
C	4.00
G	1.50
X	2.50
X1	6.50
Y	1.70

SMB	
Dimensions	**Value (in mm)**
C	4.30
G	1.80
X	2.50
X1	6.80
Y	2.30

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PACKAGE DIMENSIONS in millimeters (inches): **SOD-123**

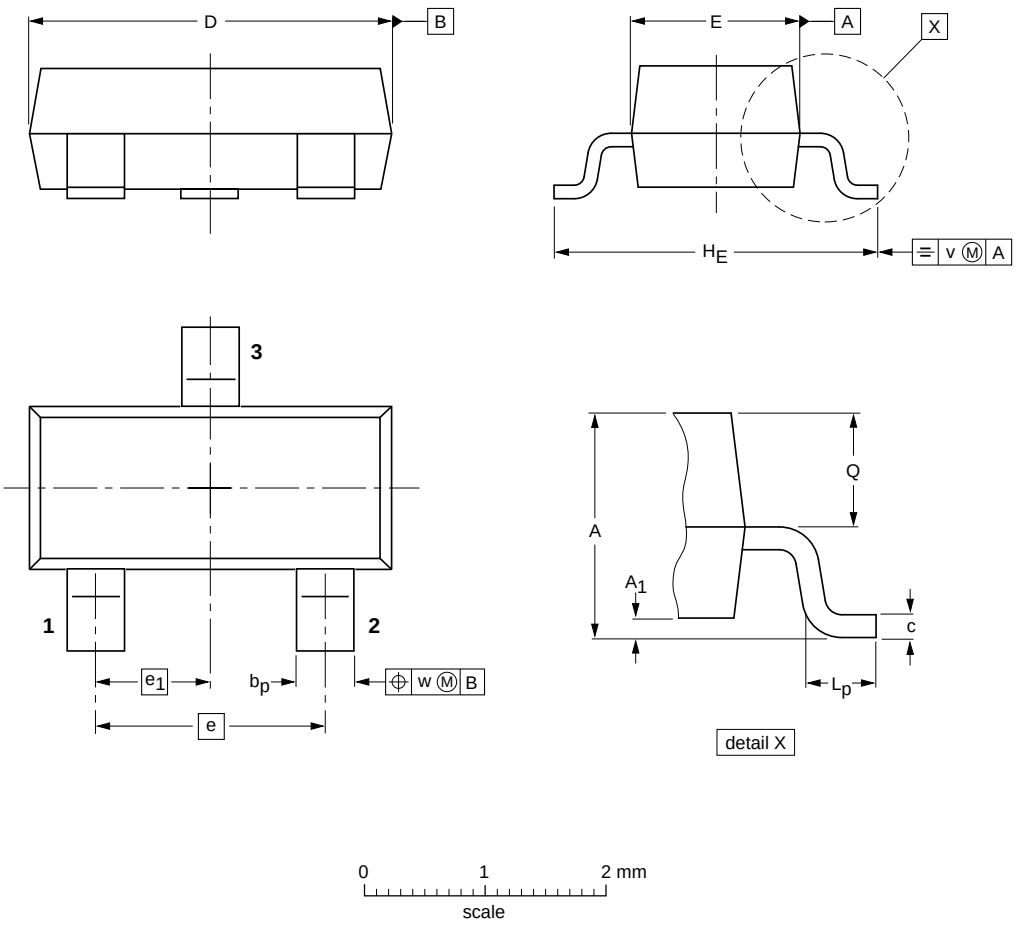


Rev. 4 - Date: 24. Sep. 2009
Document no.: S8-V-3910.01-001 (4)
17432

For technical questions within your region: DiodesAmericas@vishay.com, DiodesAsia@vishay.com, DiodesEurope@vishay.com
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9. Package outline

Plastic surface-mounted package; 3 leads **SOT23**



DIMENSIONS (mm are the original dimensions)

UNIT	A	A_1 max.	b_p	c	D	E	e	e_1	H_E	L_p	Q	v	w
mm	1.1 0.9	0.1	0.48 0.38	0.15 0.09	3.0 2.8	1.4 1.2	1.9	0.95	2.5 2.1	0.45 0.15	0.55 0.45	0.2	0.1

OUTLINE VERSION	REFERENCES				EUROPEAN PROJECTION	ISSUE DATE
	IEC	JEDEC	JEITA			
SOT23		TO-236AB				~~04-11-04~~ 06-03-16

Fig 18. Package outline SOT23 (TO-236AB)

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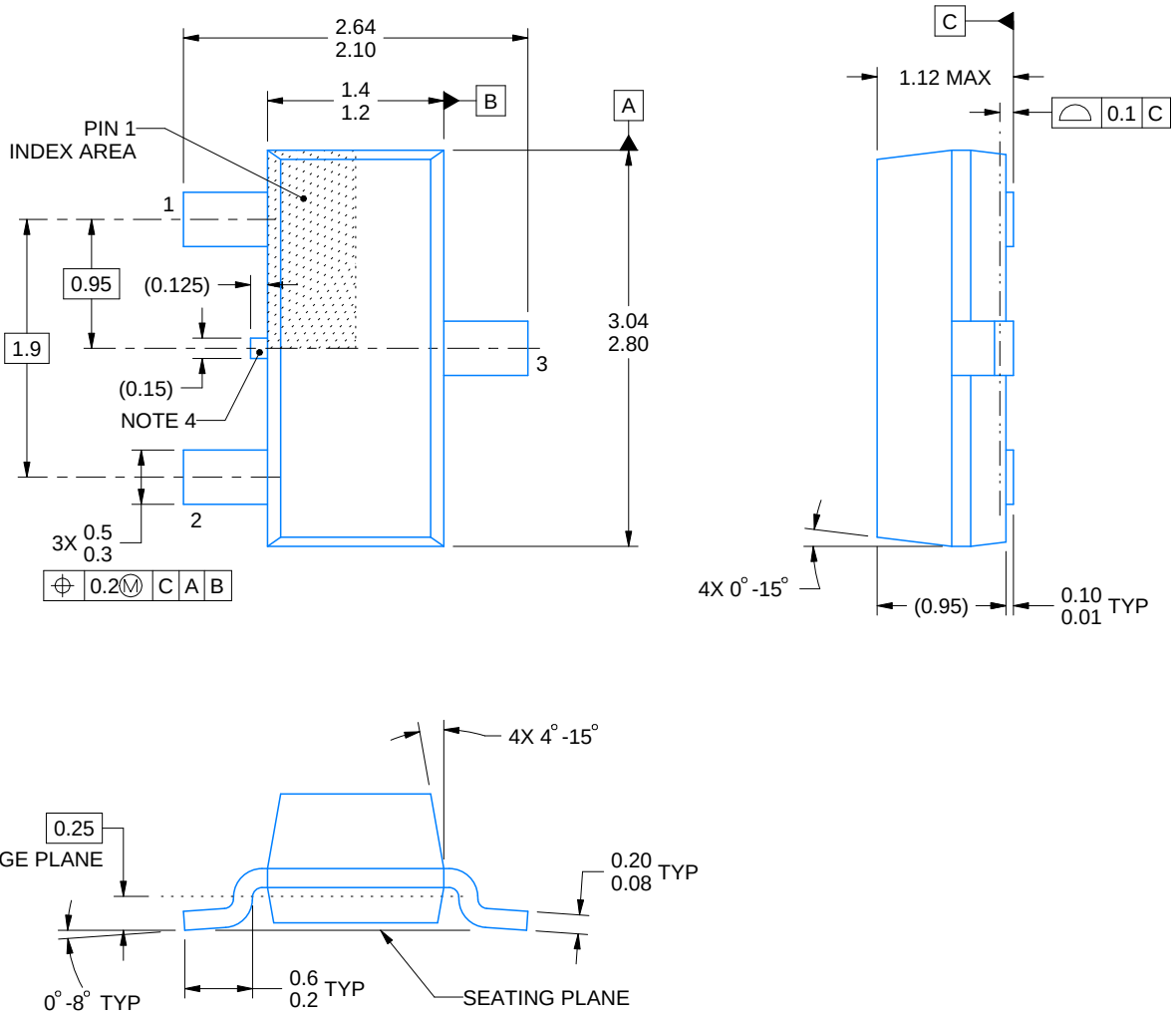
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PACKAGE OUTLINE

DBZ0003A

SOT-23 - 1.12 mm max height

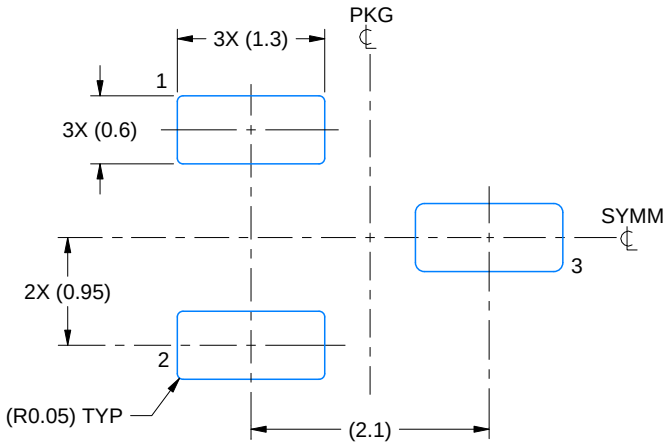
SMALL OUTLINE TRANSISTOR



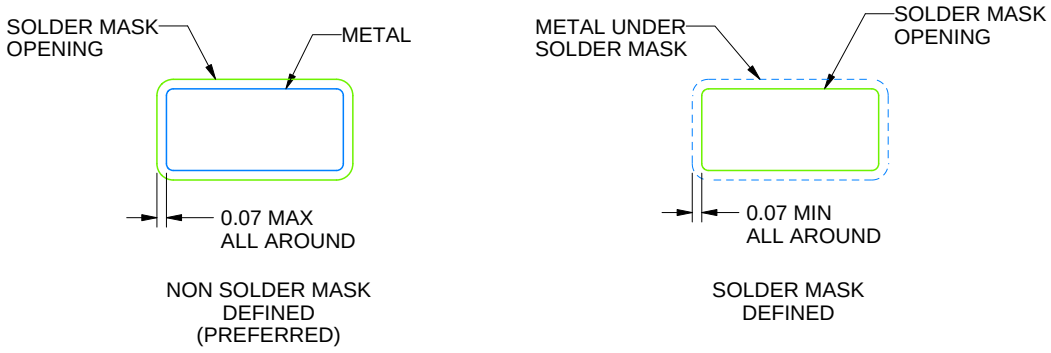
4214838/F 08/2024

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Reference JEDEC registration TO-236, except minimum foot length.
4. Support pin may differ or may not be present.
5. Body dimensions do not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.25mm per side



LAND PATTERN EXAMPLE
SCALE:15X

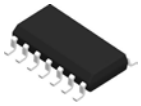


SOLDER MASK DETAILS

4214838/F 08/2024

NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

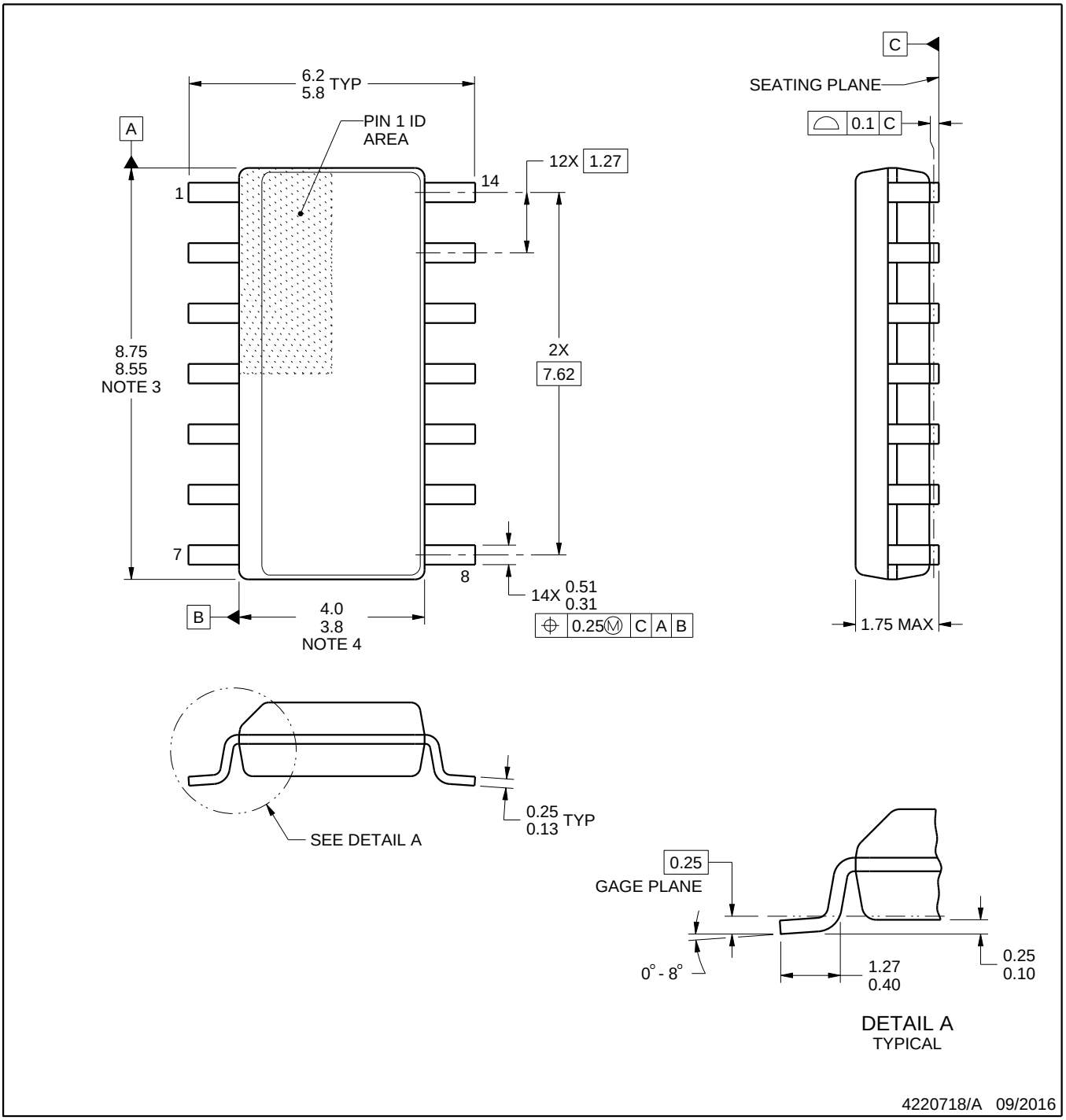


PACKAGE OUTLINE

D0014A

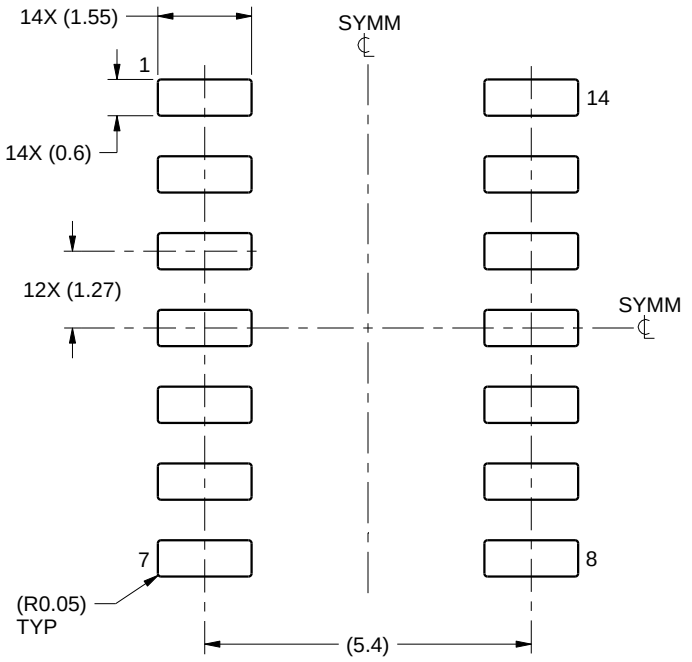
SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT

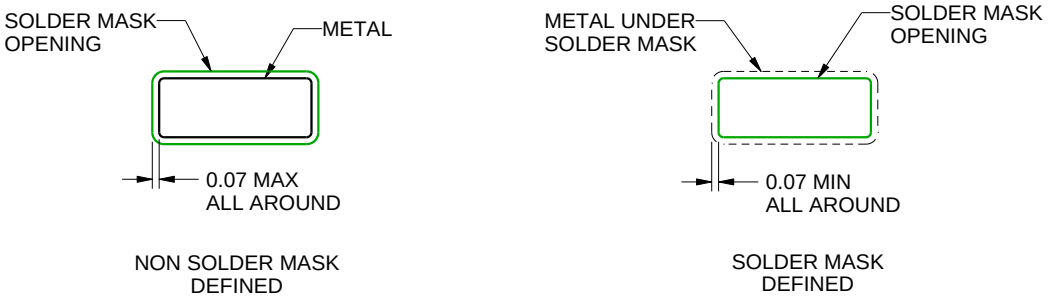


NOTES:

1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.43 mm, per side.
5. Reference JEDEC registration MS-012, variation AB.



LAND PATTERN EXAMPLE
SCALE:8X



SOLDER MASK DETAILS

4220718/A 09/2016

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

5 Device Comparison Table

Orderable Part Number	Frequency	PFM OR FPWM	Output
LMR51430XDDCR	500 kHz	PFM	Adjustable
LMR51430XFDDCR	500 kHz	FPWM	Adjustable
LMR51430YDDCR	1.1 MHz	PFM	Adjustable
LMR51430YFDDCR	1.1 MHz	FPWM	Adjustable

6 Pin Configuration and Functions

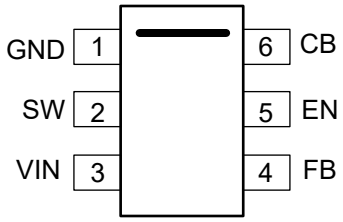


Figure 6-1. 6-Pin SOT-23 DBV Package (Top View)

Table 6-1. Pin Functions

Pin		Type[1]	Description
Name	NO.		
GND	1	G	Power ground terminals. This pin connects to the source of the low-side FET internally. Connect to system ground, and the ground side of CIN and COUT. The path to CIN must be as short as possible.
SW	2	P	Switching output of the converter. This pin connects to the source of the high-side FET and drain of the low-side FET. Connect this pin to the power inductor.
VIN	3	P	Supply input terminal to internal bias LDO and high-side FET. Connect this pin to the input supply and input bypass capacitors, C_{IN}. Input bypass capacitors must be directly connected to this pin and GND.
FB	4	A	Feedback input to the converter. Connect a resistor divider to set the output voltage. Never short this pin to ground during operation.
EN	5	A	Precision enable input to the converter. Do not float. High = on, low = off. This pin can be tied to VIN. Precision enable input allows adjustable UVLO by an external resistor divider. If the EN pin is left floating, the device is disabled.
CB	6	P	Bootstrap capacitor connection for the high-side FET driver. Connect a high quality 100-nF capacitor from this pin to the SW pin.

(1) A = Analog, P = Power, G = Ground

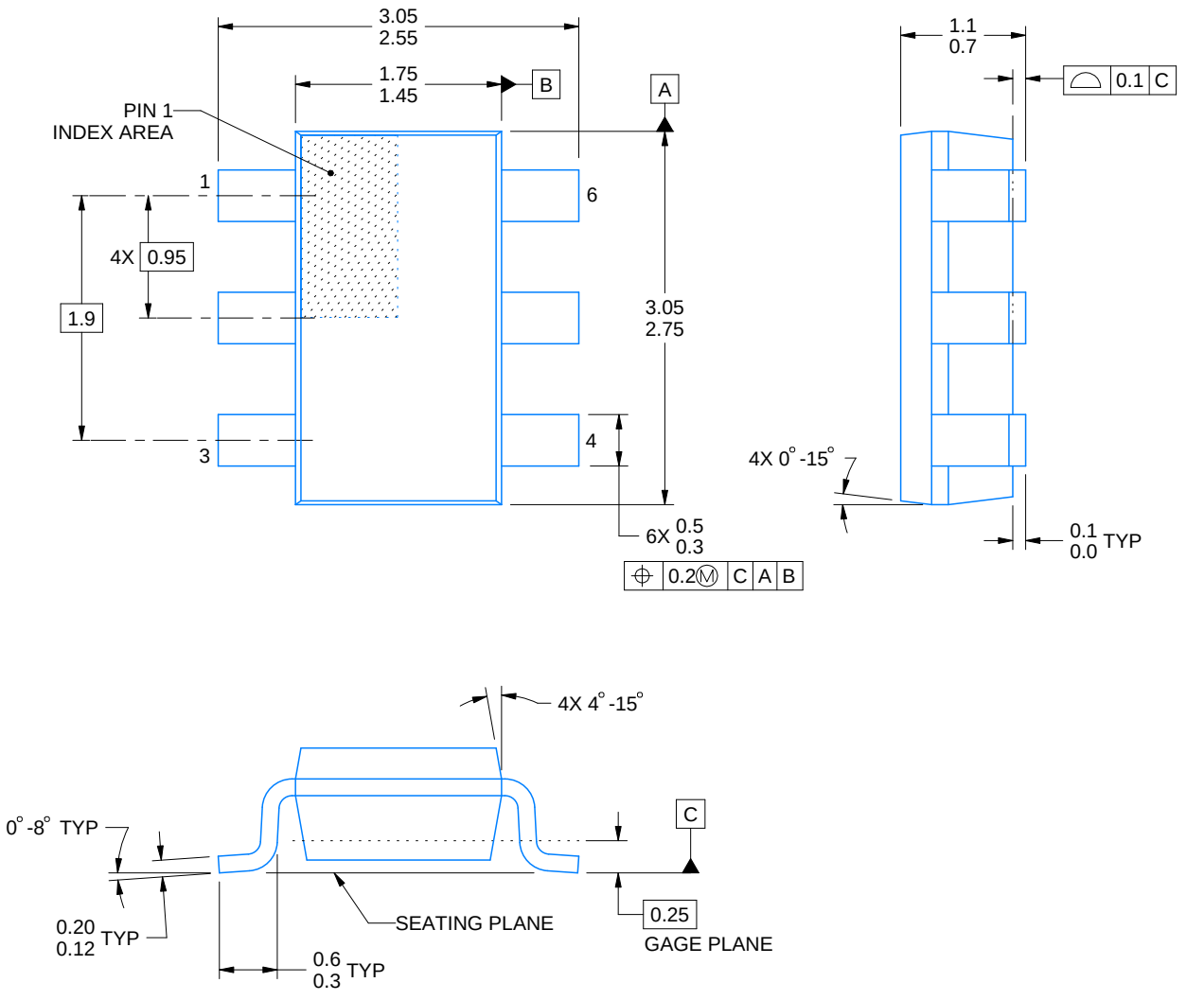
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PACKAGE OUTLINE

DDC0006A

SOT-23 - 1.1 max height

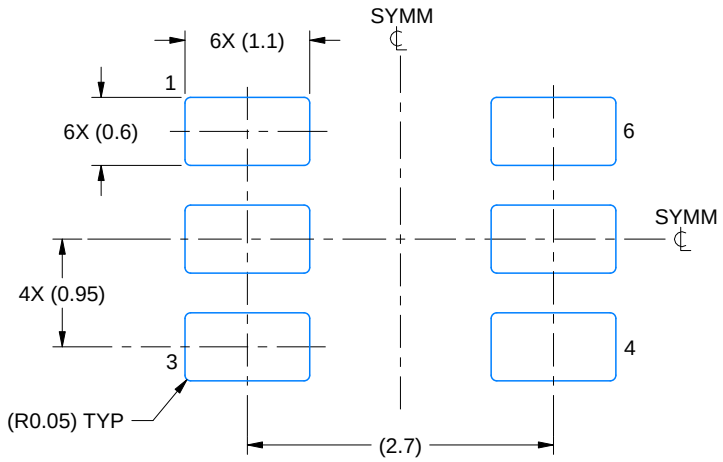
SMALL OUTLINE TRANSISTOR



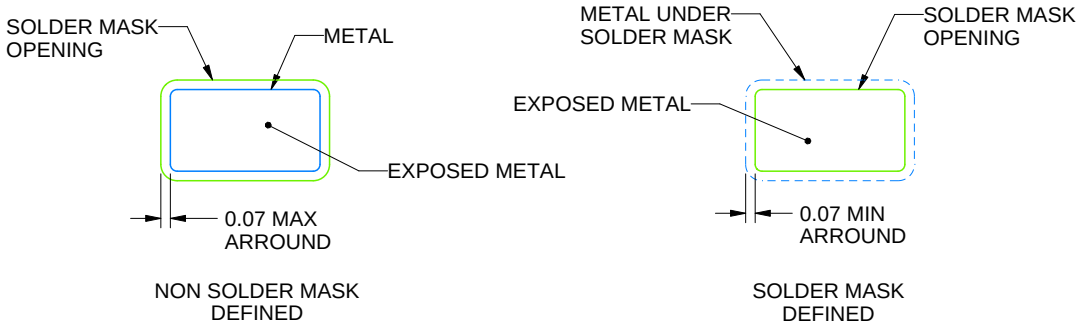
4214841/E 08/2024

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Reference JEDEC MO-193.



LAND PATTERN EXAMPLE
EXPLOSED METAL SHOWN
SCALE:15X



SOLDERMASK DETAILS

4214841/E 08/2024

NOTES: (continued)

4. Publication IPC-7351 may have alternate designs.
5. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

MECHANICAL CASE OUTLINE
PACKAGE DIMENSIONS

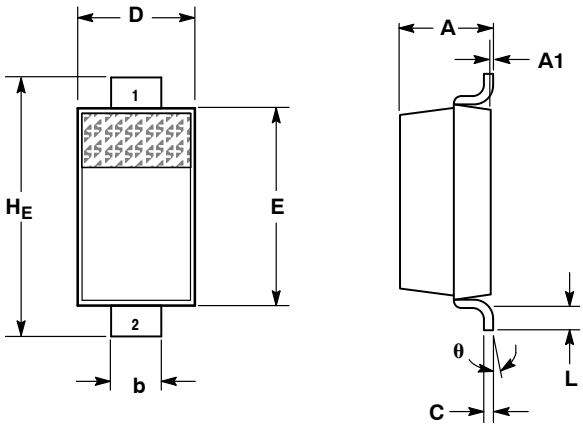
ON Semiconductor®



SCALE 5:1

SOD−123
CASE 425−04
ISSUE G

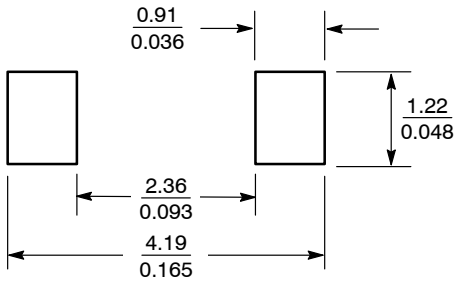
DATE 07 OCT 2009



NOTES:
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCH.

	MILLIMETERS			INCHES		
DIM	MIN	NOM	MAX	MIN	NOM	MAX
A	0.94	1.17	1.35	0.037	0.046	0.053
A1	0.00	0.05	0.10	0.000	0.002	0.004
b	0.51	0.61	0.71	0.020	0.024	0.028
c	---	---	0.15	---	---	0.006
D	1.40	1.60	1.80	0.055	0.063	0.071
E	2.54	2.69	2.84	0.100	0.106	0.112
H_E	3.56	3.68	3.86	0.140	0.145	0.152
L	0.25	---	---	0.010	---	---
θ	0°	---	10°	0°	---	10°

SOLDERING FOOTPRINT*



SCALE 10:1 $\left(\frac{\text{mm}}{\text{inches}}\right)$

*For additional information on our Pb−Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

GENERIC MARKING DIAGRAM*



XXX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

(Note: Microdot may be in either location)

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “ ▪”, may or may not be present.

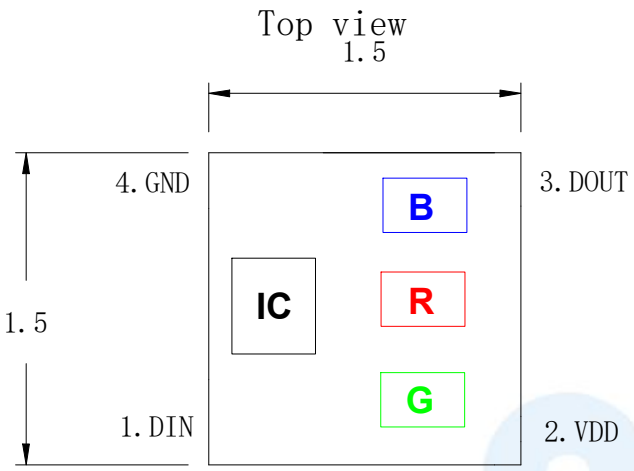
STYLE 1:
PIN 1. CATHODE
2. ANODE

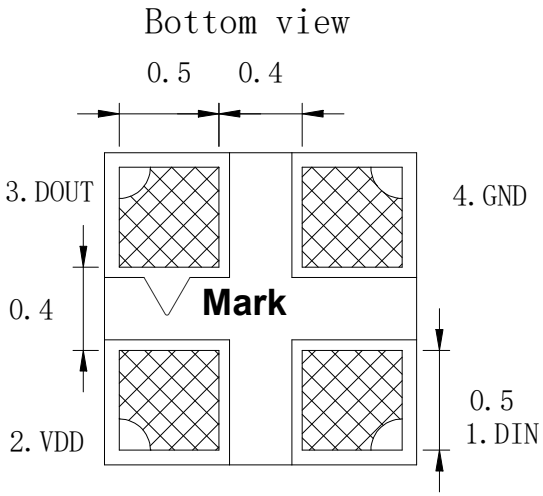
DOCUMENT NUMBER:	98ASB42927B	Electronic versions are uncontrolled except when accessed directly from the Document Repository. Printed versions are uncontrolled except when stamped “CONTROLLED COPY” in red.	
DESCRIPTION:	SOD−123		PAGE 1 OF 1

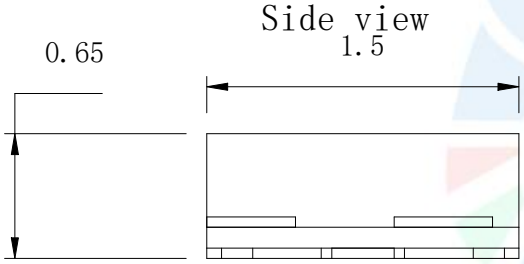
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4.机械尺寸 Mechanical dimensions:





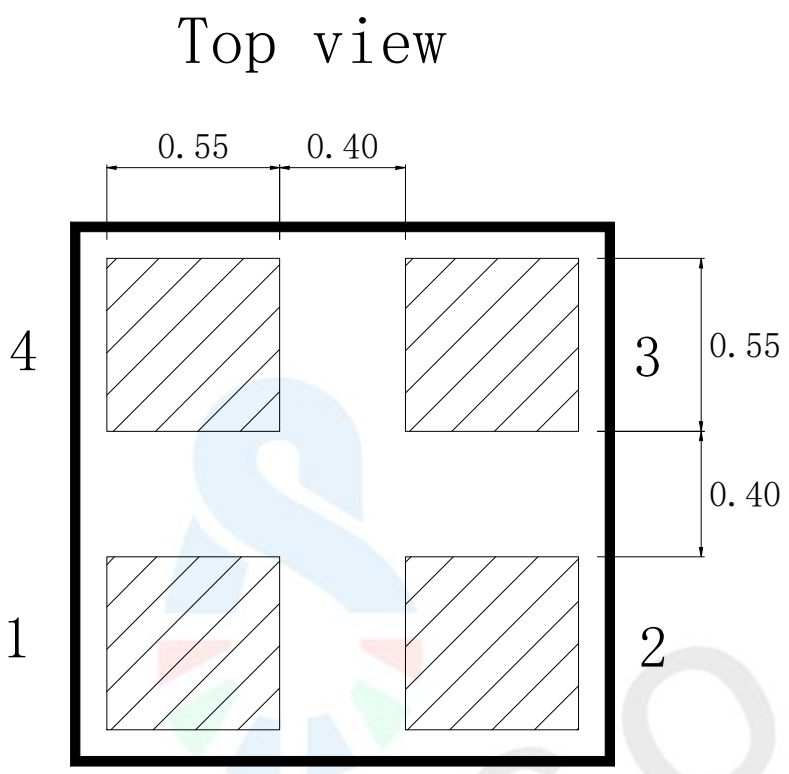


备注 Notes:
1. 以上标示单位为毫米.
The above markings are in millimeters
2. 除非另外注明，尺寸公差为 ±0.1毫米.
Unless otherwise specified, the dimensional tolerance is ± 0.1 millimeters

5. 引脚功能说明 Pin Function Description:

序号 Serial Number	符号 Symbol	管脚名 Pin name	功能描述 Function Description
1	DIN	数据输入 Data input	控制数据信号输入 Control data signal input
2	VDD	电源 Power supply	供电管脚 Power supply pins
3	DOUT	数据输出 Data output	控制数据信号输出 Control data signal output
4	GND	地 Grounds	电源接地 Power grounding

6.PCB建议焊盘尺寸 PCB recommended pad size：(以实际使用情况为准 Based on actual usage)



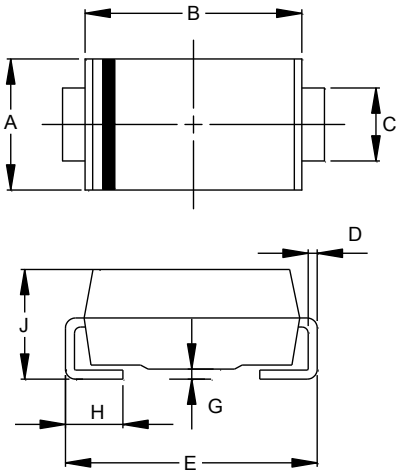
7.IC极限参数 IC limit parameter： (Ta=25℃)

参数 Parameter	符号 Symbol	范围 Range	单位 Unit
逻辑电源电压 Working voltage	VDD	3.7-5.5	V
工作温度 Operation temperature	Topt	-40~+80	℃
储存温度 Storage temperature	Tstg	-40~+80	℃
ESD耐压（人体模式） ESD withstand voltage (human mode)	V_{ESD}	2K	V

Package Outline Dimensions

Please see http://www.diodes.com/package-outlines.html for the latest version.

SMA

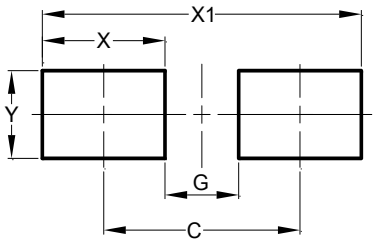


SMA		
Dim	**Min**	**Max**
A	2.29	2.92
B	4.00	4.60
C	1.27	1.63
D	0.15	0.31
E	4.80	5.59
G	0.05	0.20
H	0.76	1.52
J	1.96	2.40
All Dimensions in mm		

Suggested Pad Layout

Please see http://www.diodes.com/package-outlines.html for the latest version.

SMA



Dimensions	Value (in mm)
C	4.00
G	1.50
X	2.50
X1	6.50
Y	1.70

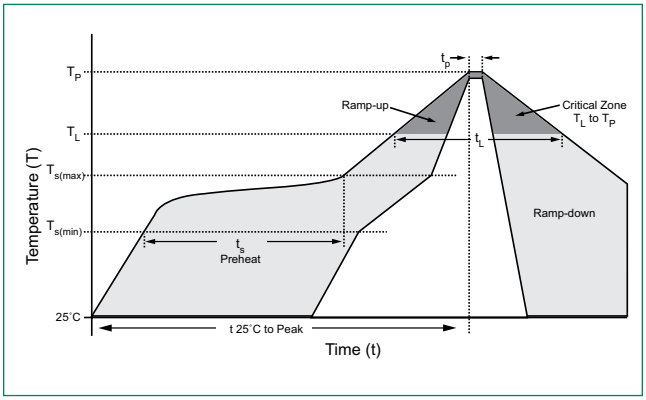
© Diodes Incorporated

Transient Voltage Suppression Diodes

Surface Mount – 400W > SMAJ series

Soldering Parameters

Reflow Condition		Lead–free assembly
Pre Heat	-Temperature Min ($T_{s(min)}$)	150°C
	-Temperature Max ($T_{s(max)}$)	200°C
	-Time (min to max) (t_s)	60 – 180 secs
Average ramp up rate (Liquidus Temp (T_A) to peak		3°C/second max
$T_{S(max)}$ to T_A - Ramp-up Rate		3°C/second max
Reflow	-Temperature (T_A) (Liquidus)	217°C
	-Time (min to max) (t_s)	60 – 150 seconds
Peak Temperature (T_P)		$260^{+0/-5}$ °C
Time within 5°C of actual peak Temperature (t_p)		20 – 40 seconds
Ramp-down Rate		6°C/second max
Time 25°C to peak Temperature (T_P)		8 minutes Max.
Do not exceed		260°C



Physical Specifications

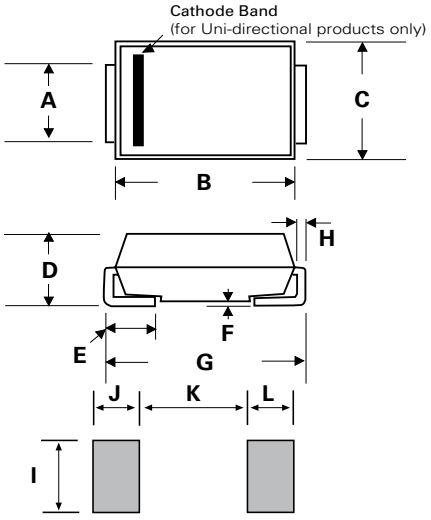
Weight	0.002 ounce, 0.061 gram
Case	JEDEC DO-214AC Molded Plastic over glass passivated junction
Polarity	Color band denotes cathode except Bipolar
Terminal	Matte Tin-plated leads, Solderable per JESD22-B102

Environmental Specifications

High Temp. Storage	JESD22-A103
HTRB	JESD22-A108
Temperature Cycling	JESD22-A104
MSL	JEDEC-J-STD-020, Level 1
H3TRB	JESD22-A101
RSH	JESD22-A111

Dimensions

DO-214AC (SMA)

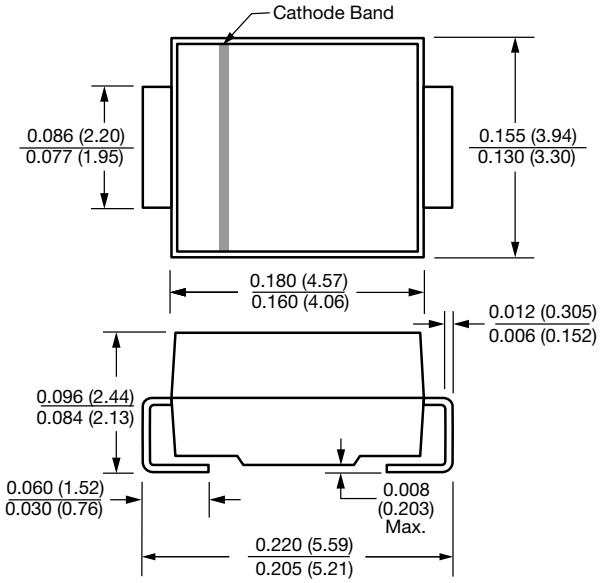


Dimensions	Inches		Millimeters	
	Min	Max	Min	Max
A	0.049	0.065	1.250	1.650
B	0.157	0.181	3.990	4.600
C	0.095	0.110	2.400	2.790
D	0.075	0.090	1.900	2.290
E	0.030	0.060	0.780	1.520
F	-	0.008	-	0.203
G	0.189	0.208	4.800	5.280
H	0.006	0.012	0.152	0.305
I	0.070	-	1.800	-
J	0.082	-	2.100	-
K	-	0.090	-	2.300
L	0.082	-	2.100	-

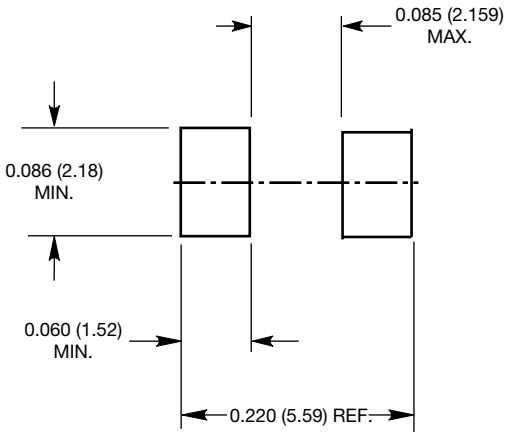
© 2015 Littelfuse, Inc.
Specifications are subject to change without notice.
Revised: 11/20/15

PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

SMB (DO-214AA)



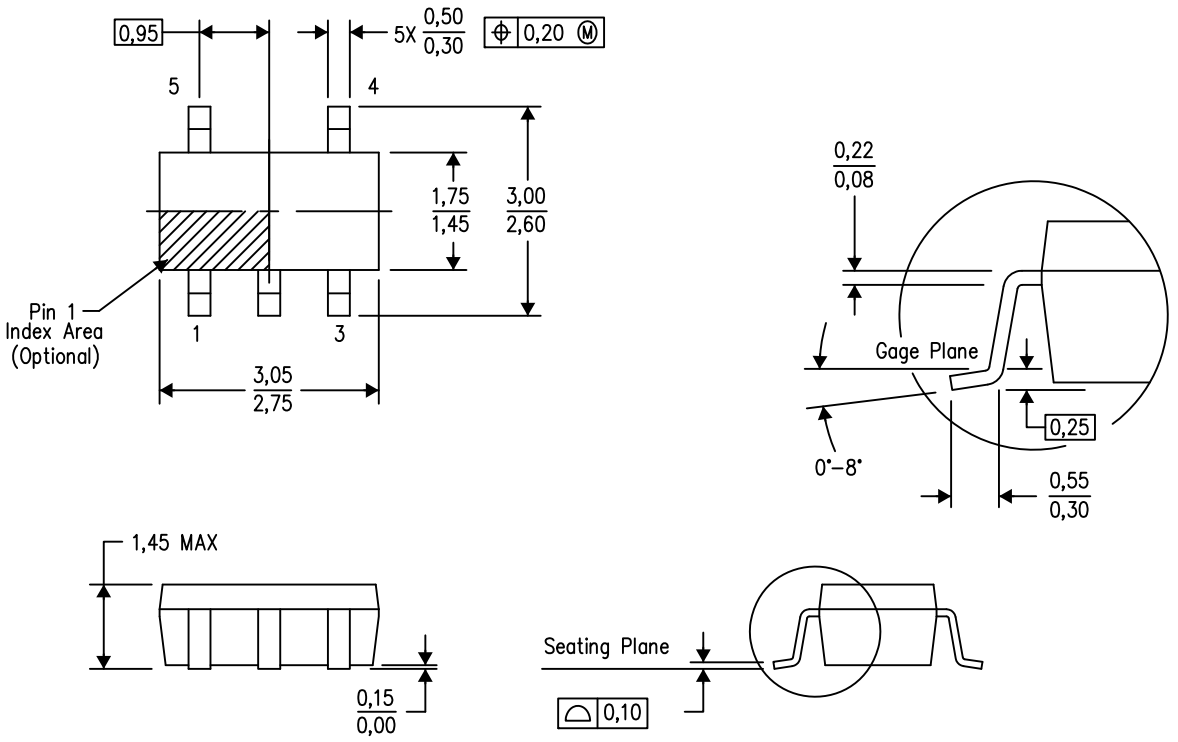
Mounting Pad Layout



For technical questions within your region: DiodesAmericas@vishay.com, DiodesAsia@vishay.com, DiodesEurope@vishay.com

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DBV (R–PDSO–G5) PLASTIC SMALL–OUTLINE PACKAGE



4073253–4/L 08/2013

NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusion. Mold flash and protrusion shall not exceed 0.15 per side.
D. Falls within JEDEC MO–178 Variation AA.



Features

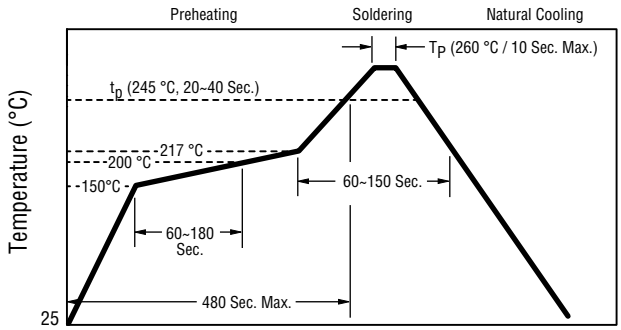
- Shielded construction
- Carbonyl powder core
- High saturation current
- Inductance range: 0.1 to 10 µH
- AEC-Q200 compliant
- RoHS compliant* and halogen free**

SRP5030TA Series - Shielded Power Inductors

Electrical Specifications @ 25 °C

Bourns Part Number	Inductance @ 100 KHz / 1 V		Q (Min.) @100 KHz / 1 V	SRF (MHz) Typ.	DCR (mΩ) Typ.	DCR (mΩ) Max.	Irms (A)	Isat (A)	Terminal Type
	L (µH)	Tol. (%)							
SRP5030TA-R10Y	0.1	± 30	5	320	2.5	3	23	27	Lead Frame
SRP5030TA-R22M	0.22	± 20	5	155	3.7	4.4	15.5	21	
SRP5030TA-R33M	0.33	± 20	5	110	4.3	5	14	18	
SRP5030TA-R47M	0.47	± 20	10	100	6.4	7.4	12	16	
SRP5030TA-R68M	0.68	± 20	10	85	10	12	8.5	14	
SRP5030TA-1R0M	1.0	± 20	10	62	13	14	7	11	
SRP5030TA-1R2M	1.2	± 20	10	50	14	16	6.5	11	
SRP5030TA-1R5M	1.5	± 20	10	44	16	25	6	10	
SRP5030TA-2R2M	2.2	± 20	15	36	25	35	5.5	9	
SRP5030TA-3R3M	3.3	± 20	15	30	32	38	5	8	
SRP5030TA-3R6M	3.6	± 20	15	29	46	54	4.9	7	
SRP5030TA-4R7M	4.7	± 20	15	23	50	53	4.6	6	
SRP5030TA-5R6M	5.6	± 20	15	21	55	63	4.25	4.5	
SRP5030TA-6R8M	6.8	± 20	15	19	68	76.2	4	4.3	
SRP5030TA-100M	10	± 20	20	18	110	128	2.75	3.5	

Soldering Profile



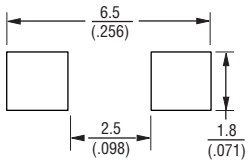
Reflow Times: 3 Times Max.

How to Order

SRP5030TA - 100 M

Model
Value Code (see table)
Tolerance Code

Recommended Layout



WARNING Cancer and Reproductive Harm www.P65Warnings.ca.gov

* RoHS Directive 2015/863, Mar 31, 2015 and Annex.

** Bourns considers a product to be "halogen free" if (a) the Bromine (Br) content is 900 ppm or less; (b) the Chlorine (Cl) content is 900 ppm or less; and (c) the total Bromine (Br) and Chlorine (Cl) content is 1500 ppm or less.

Specifications are subject to change without notice. Users should verify actual device performance in their specific applications. The products described herein and this document are subject to specific legal disclaimers as set forth on the last page of this document, and at www.bourns.com/docs/legal/disclaimer.pdf.

Additional Information

Click these links for more information:

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General Specifications

Operating Temperature
.....-55 °C to +150 °C
(Temperature rise included)
Storage Temperature
.....-55 °C to +150 °C
Rated Current
..... Inductance drops 20 % at Isat
Temperature Rise .....40 °C at rated Irms
Moisture Sensitivity Level 1
ESD Classification (HBM).....N/A

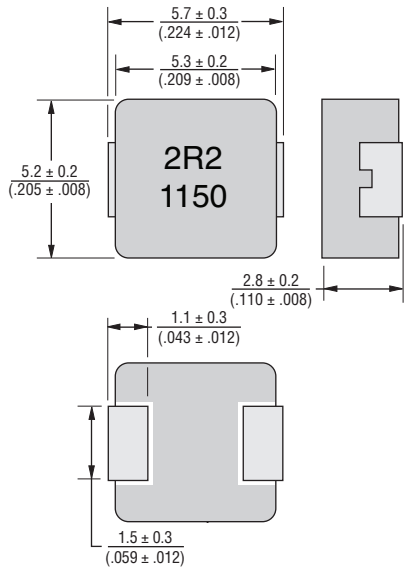
Note 1: Circuit design, component, PCB trace size and thickness, airflow and other cooling provisions all affect the part temperature. Part temperature should be verified in the end application.

Materials

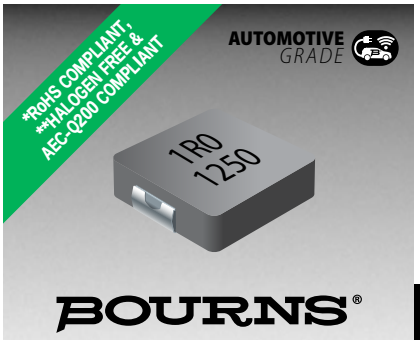
Core.....Carbonyl powder
Wire .....Enameled copper
Terminal Finish .....Sn
Packaging..... 2000 pcs. per 13-inch reel

Product Dimensions

Lead Frame Terminal



DIMENSIONS: MM (INCHES)



Features

- Shielded construction
- Carbonyl powder core
- High saturation current
- Inductance range: 0.10 to 33 µH
- AEC-Q200 compliant
- RoHS compliant* and halogen free**

SRP7028A Series - Shielded Power Inductors

Electrical Specifications @ 25 °C

Bourns Part Number	Inductance @ 100 KHz / 1 V L (µH)	Tol. (%)	Q (Min.) @100 KHz / 1 V	SRF (MHz) Typ.	DCR (mΩ) Typ.	DCR (mΩ) Max.	Irms (A)	Isat (A)	Terminal Type
SRP7028A-R10M	0.10	± 20	5	300	1.2	1.7	32.5	60	Lead Frame
SRP7028A-R15Y	0.15	± 30	5	220	1.5	1.9	27	45	
SRP7028A-R22Y	0.22	± 30	5	150	2.1	2.8	23	40	
SRP7028A-R33M	0.33	± 20	10	140	3.5	3.9	20	32	
SRP7028A-R47M	0.47	± 20	10	80	4	4.2	17.5	26	
SRP7028A-R56M	0.56	± 20	15	70	4.7	5.0	16.5	25.5	
SRP7028A-R68M	0.68	± 20	15	67	4.8	5.5	15.5	25	
SRP7028A-R75M	0.75	± 20	15	65	5.5	6.6	14.5	24.5	
SRP7028A-R82M	0.82	± 20	15	61	6.7	8.0	13	24	
SRP7028A-1R0M	1.0	± 20	20	58	8.3	10.0	11	22	
SRP7028A-1R5M	1.5	± 20	20	38	13	15.0	9	18	
SRP7028A-1R8M	1.8	± 20	20	36	14	17.0	8.5	16	
SRP7028A-2R2M	2.2	± 20	20	32	18	20.0	8	14	
SRP7028A-2R5M	2.5	± 20	20	30	20	22.0	7	13	
SRP7028A-3R3M	3.3	± 20	20	25	28	30.0	6	12	
SRP7028A-4R7M	4.7	± 20	20	21	37	40.0	5.5	10	
SRP7028A-5R6M	5.6	± 20	20	19	43	48.0	5	9	
SRP7028A-6R8M	6.8	± 20	20	18	54	60.0	4.5	8	
SRP7028A-8R2M	8.2	± 20	20	17	64	68.0	4	7.5	
SRP7028A-100M	10	± 20	20	14	75	85.0	3.5	6	
SRP7028A-150M	15	± 20	20	12	107	123.0	3.0	4	
SRP7028A-220M	22	± 20	20	10	165	190.0	2.0	3.5	
SRP7028A-330M	33	± 20	20	8	200	240.0	2.0	2.5	

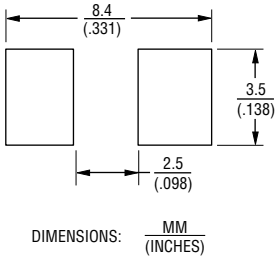
How to Order

SRP7028A - 100M

Model

Value Code (see table)

Recommended Layout



WARNING
Cancer and Reproductive Harm
www.P65Warnings.ca.gov

* RoHS Directive 2015/863, Mar 31, 2015 and Annex.

** Bourns considers a product to be "halogen free" if (a) the Bromine (Br) content is 900 ppm or less; (b) the Chlorine (Cl) content is 900 ppm or less; and (c) the total Bromine (Br) and Chlorine (Cl) content is 1500 ppm or less.

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Additional Information

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General Specifications

Test Voltage ..... 1.0 V
Test Frequency ..... 100 KHz
Operating Temperature
..... -55 °C to +150 °C
(Temperature rise included)
Storage Temperature ... -55 °C to +150 °C
Rated Current
..... Inductance drops 20 % at Isat
Temperature Rise .....40 °C at rated Irms
Moisture Sensitivity Level 1
ESD Classification (HBM).....N/A

[1] Circuit design, component, PCB trace size and thickness, airflow and other cooling provisions all affect the part temperature. Part temperature should be verified in the end application.

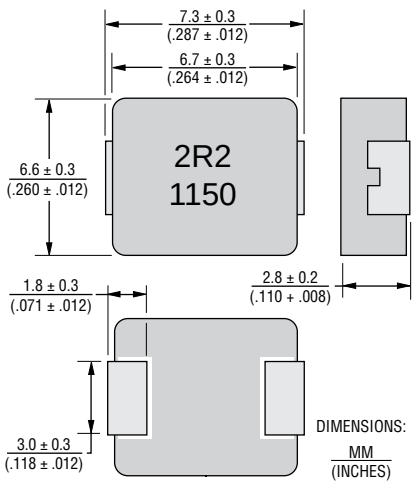
Materials

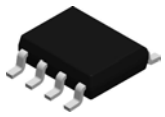
Core.....Carbonyl powder
Wire .....Enameled copper
Terminal Finish ..... Sn
Packaging..... 1000 pcs. per 13-inch reel

Please consult *Bourns® Model SRP-A Series Visual Inspection Guide* for detailed construction, processing and testing information.

Product Dimensions

Lead Frame Terminal



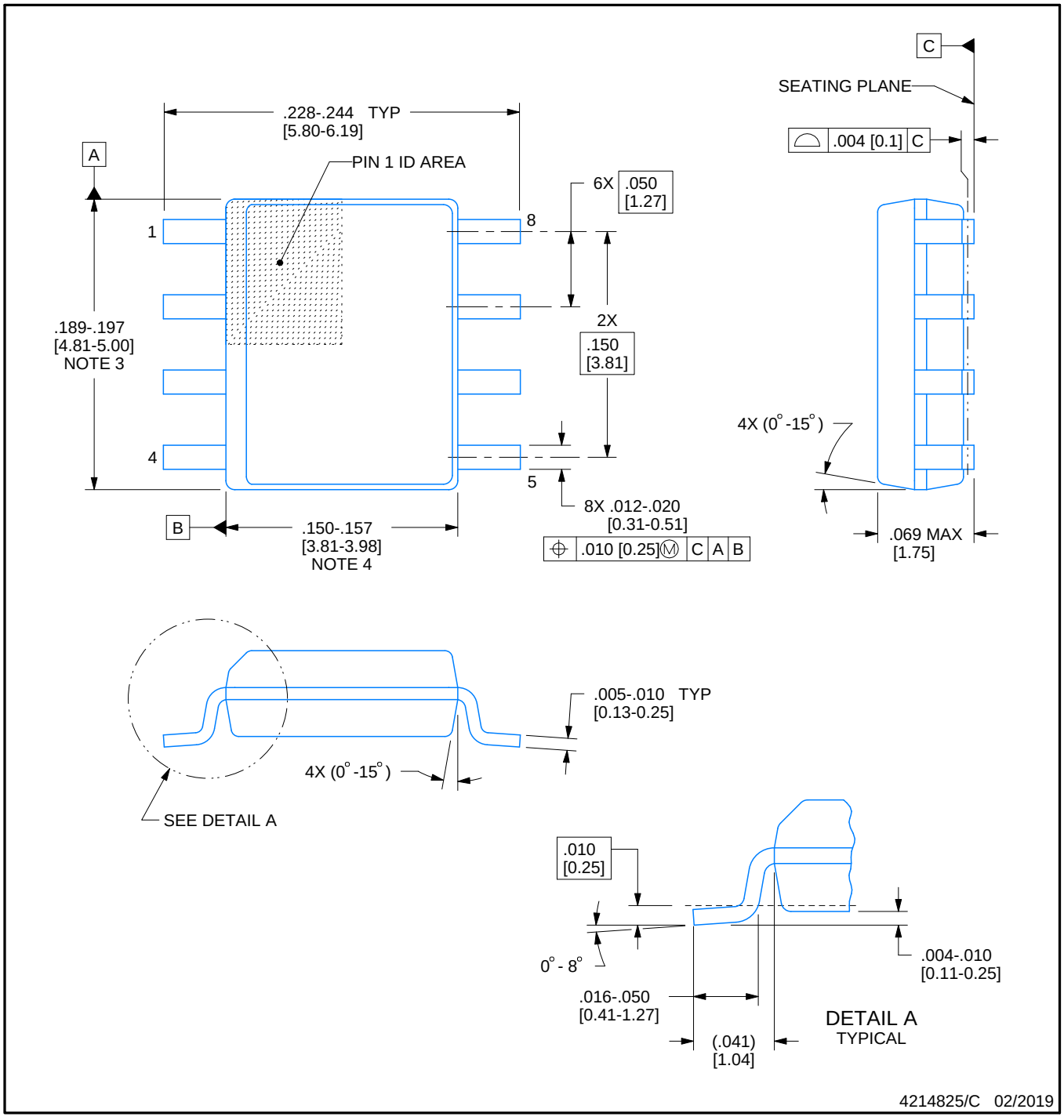


PACKAGE OUTLINE

D0008A

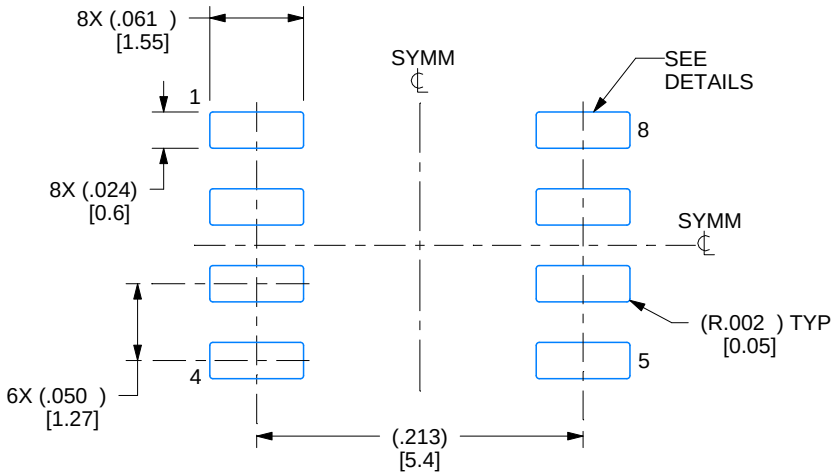
SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT

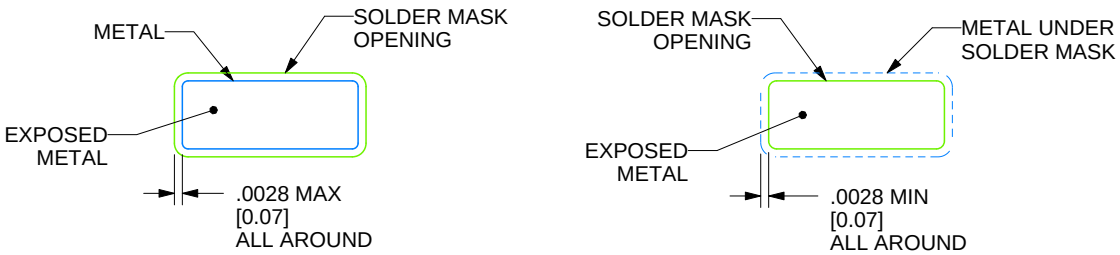


NOTES:

1. Linear dimensions are in inches [millimeters]. Dimensions in parenthesis are for reference only. Controlling dimensions are in inches. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed .006 [0.15] per side.
4. This dimension does not include interlead flash.
5. Reference JEDEC registration MS-012, variation AA.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:8X



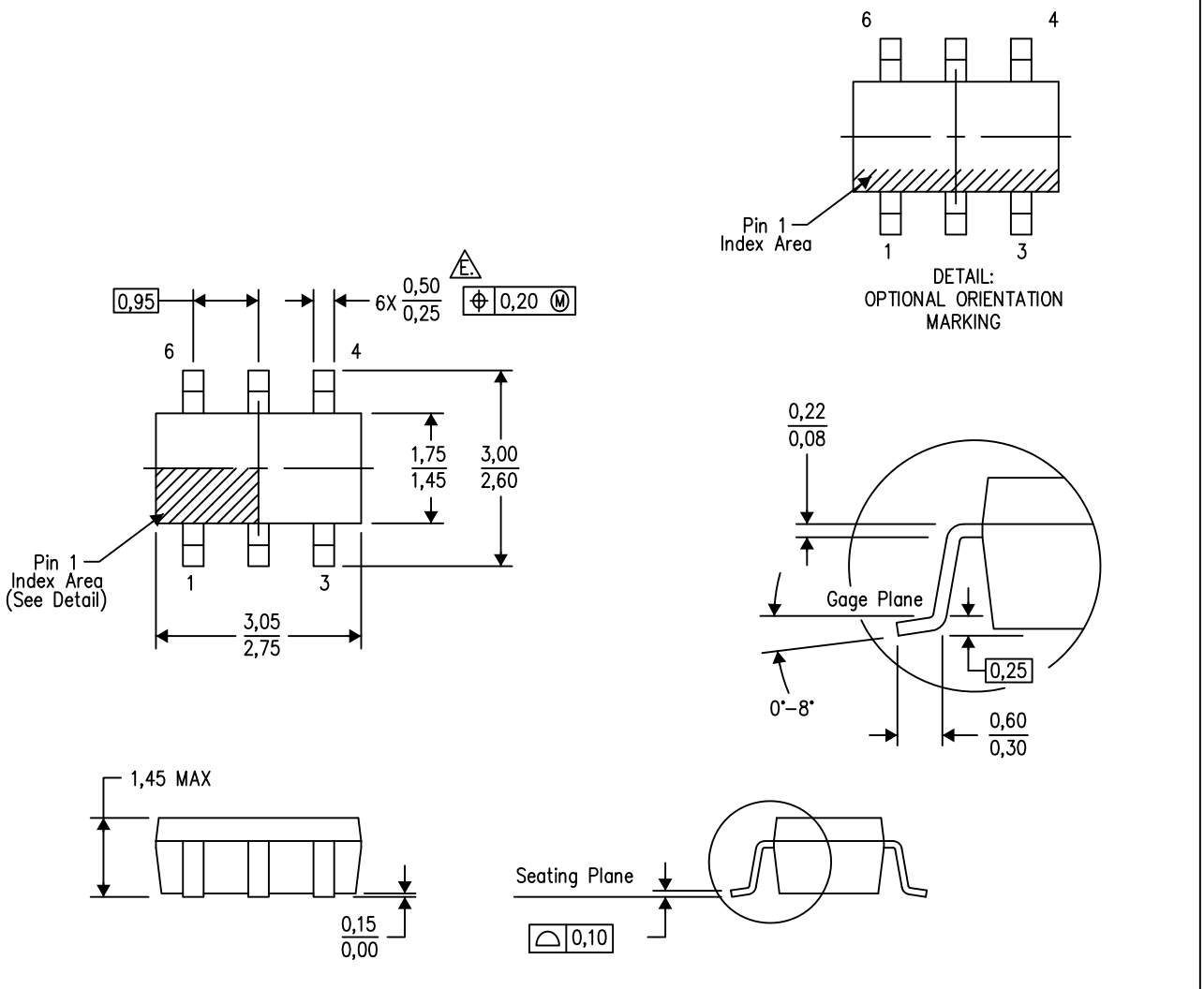
SOLDER MASK DETAILS

4214825/C 02/2019

NOTES: (continued)

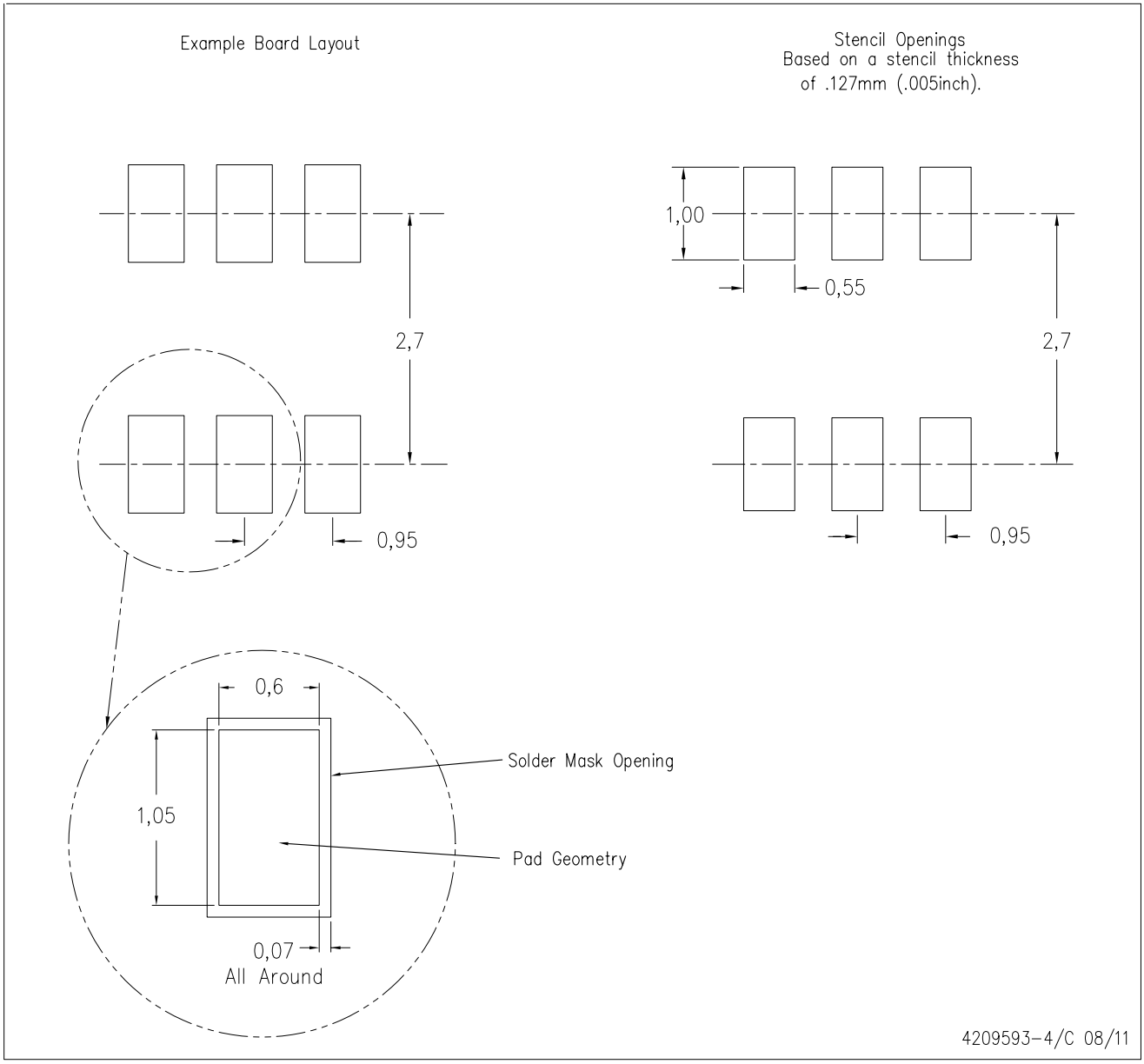
6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

DBV (R–PDSO–G6) PLASTIC SMALL–OUTLINE PACKAGE



NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusion. Mold flash and protrusion shall not exceed 0.15 per side.
D. Leads 1,2,3 may be wider than leads 4,5,6 for package orientation.
E. Falls within JEDEC MO–178 Variation AB, except minimum lead width.

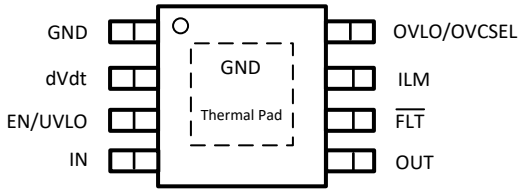
DBV (R–PDSO–G6) PLASTIC SMALL OUTLINE



NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Customers should place a note on the circuit board fabrication drawing not to alter the center solder mask defined pad.
D. Publication IPC–7351 is recommended for alternate designs.
E. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Example stencil design based on a 50% volumetric metal load solder paste. Refer to IPC–7525 for other stencil recommendations.

6 Pin Configuration and Functions

DDA Package
8-Pin SOIC
Top View

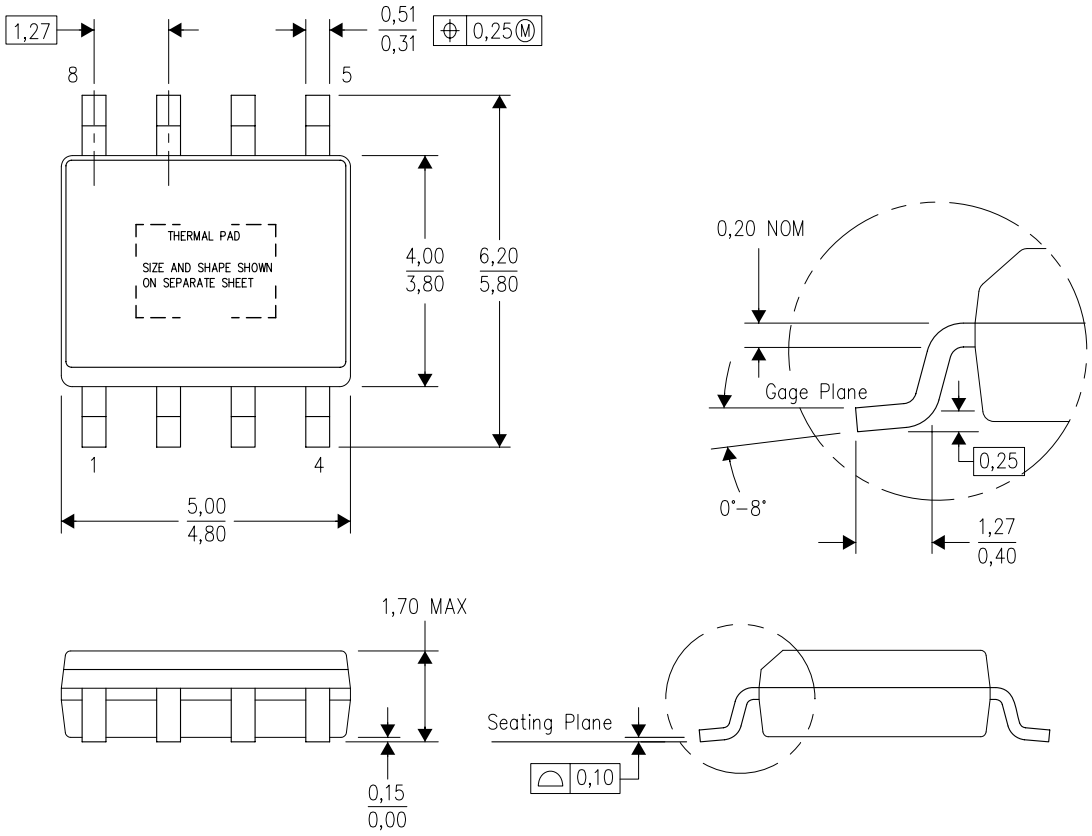


Pin Functions

PIN		I/O	DESCRIPTION
NAME	NO.		
GND	1	Ground	Ground
dVdt	2	Analog Output	A capacitor from this pin to GND sets the output turn on slew rate. Leave this pin floating for the fastest turn on slew rate.
EN/UVLO	3	Analog Input	Active High Enable for the Device. A resistor divider can be used to adjust the Undervoltage Lockout threshold. **Do not leave floating.**
IN	4	Power	Power Input
OUT	5	Power	Power Output
$\overline{\text{FLT}}$	6	Digital Output	Active Low indicator which will be pulled low when a fault is detected. It is an open-drain output that requires an external pull-up resistance.
ILM	7	Analog Output	This is a dual function pin used to limit and monitor the output current. An external resistor from this pin to GND sets the output current limit. The pin voltage can also be used to monitor the output load current.
OVLO	8	Analog Input	**TPS25963x**: A resistor divider can be used to adjust the Overvoltage Lockout threshold. **Do not leave floating.**
OVCSEL			**TPS25962x:** Overvoltage Clamp level select pin. Refer to *Overvoltage Clamp* for more details.
Thermal pad		Ground	The Exposed Pad is used primarily for heat dissipation and must be connected to system ground plane for best thermal performance.

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DDA (R–PDSO–G8) PowerPAD™ PLASTIC SMALL–OUTLINE



NOTES:
A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5–1994.
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusion not to exceed 0,15.
D. This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPad Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 for information regarding recommended board layout. This document is available at www.ti.com <http://www.ti.com>.
E. See the additional figure in the Product Data Sheet for details regarding the exposed thermal pad features and dimensions.
F. This package complies to JEDEC MS–012 variation BA

PowerPAD is a trademark of Texas Instruments.

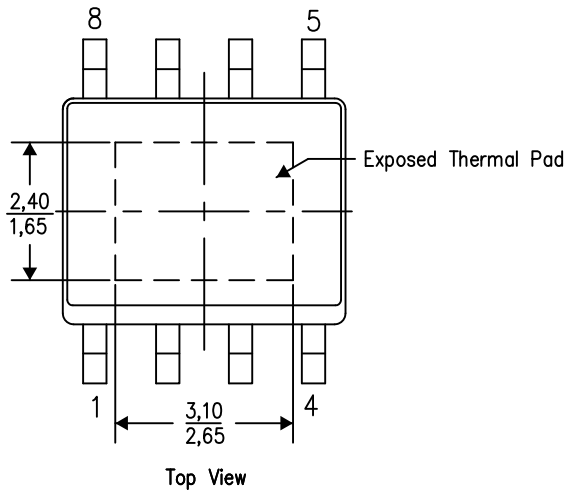
DDA (R–PDSO–G8) PowerPAD™ PLASTIC SMALL OUTLINE

THERMAL INFORMATION

This PowerPAD™ package incorporates an exposed thermal pad that is designed to be attached to a printed circuit board (PCB). The thermal pad must be soldered directly to the PCB. After soldering, the PCB can be used as a heatsink. In addition, through the use of thermal vias, the thermal pad can be attached directly to the appropriate copper plane shown in the electrical schematic for the device, or alternatively, can be attached to a special heatsink structure designed into the PCB. This design optimizes the heat transfer from the integrated circuit (IC).

For additional information on the PowerPAD package and how to take advantage of its heat dissipating abilities, refer to Technical Brief, PowerPAD Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 and Application Brief, PowerPAD Made Easy, Texas Instruments Literature No. SLMA004. Both documents are available at www.ti.com.

The exposed thermal pad dimensions for this package are shown in the following illustration.



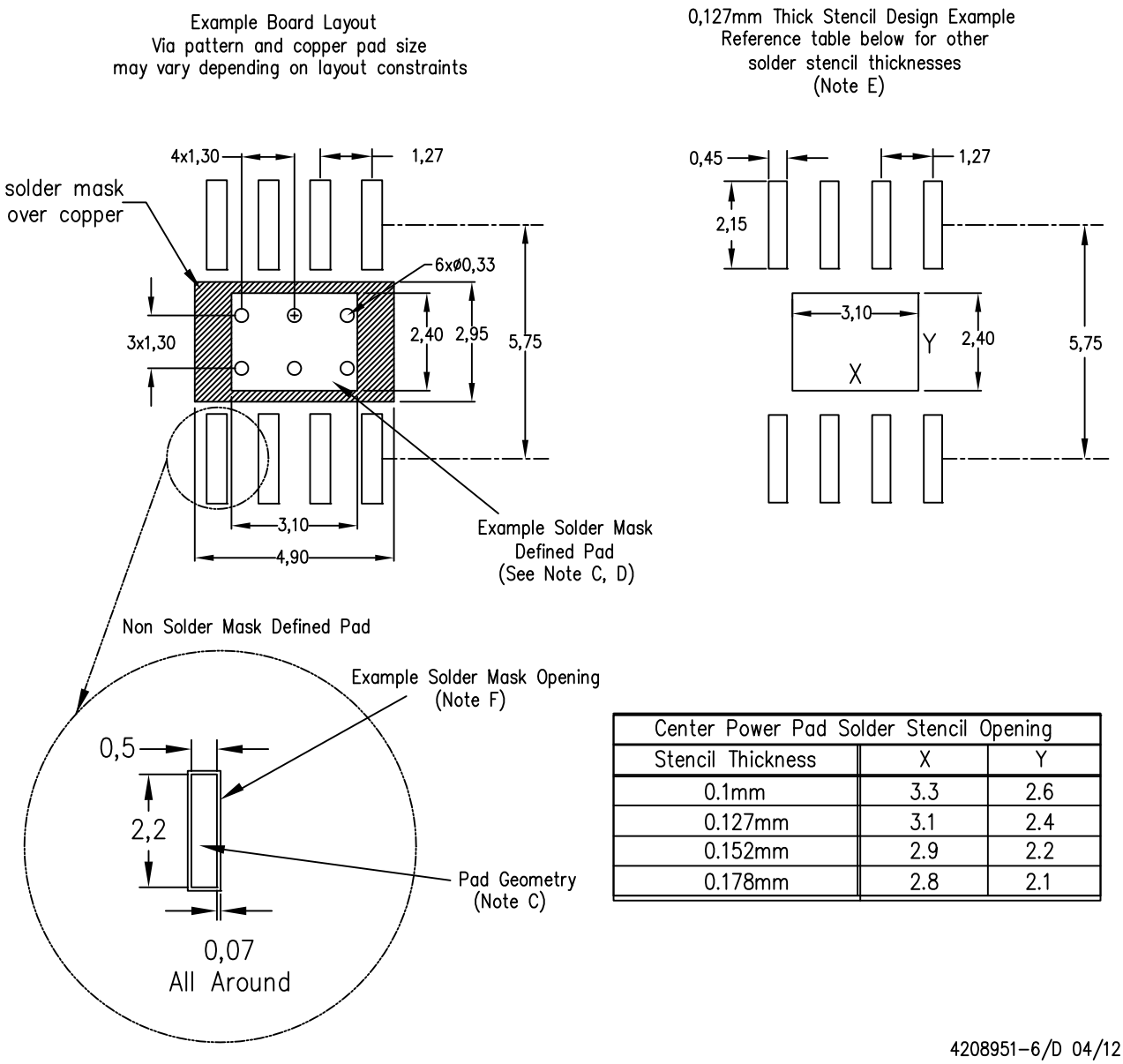
Exposed Thermal Pad Dimensions

4206322–6/L 05/12

NOTE: A. All linear dimensions are in millimeters

PowerPAD is a trademark of Texas Instruments

DDA (R–PDSO–G8) PowerPAD™ PLASTIC SMALL OUTLINE



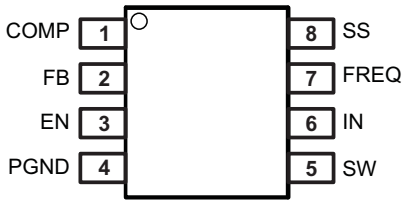
Center Power Pad Solder Stencil Opening		
Stencil Thickness	X	Y
0.1mm	3.3	2.6
0.127mm	3.1	2.4
0.152mm	2.9	2.2
0.178mm	2.8	2.1

NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Publication IPC–7351 is recommended for alternate designs.
D. This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPad Thermally Enhanced Package, Texas Instruments Literature No. SLMA002, SLMA004, and also the Product Data Sheets for specific thermal information, via requirements, and recommended board layout. These documents are available at www.ti.com <http://www.ti.com>. Publication IPC–7351 is recommended for alternate designs.
E. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Example stencil design based on a 50% volumetric metal load solder paste. Refer to IPC–7525 for other stencil recommendations.
F. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

PowerPAD is a trademark of Texas Instruments.

6 Pin Configuration and Functions

DGK, PW Packages
8 Pins
Top View



8-PIN 4.9-mm × 3-mm × 1.1-mm VSSOP (DGK)

8-PIN 6.4-mm × 3-mm × 1.2-mm TSSOP (PW)

Pin Functions

PIN		I/O	DESCRIPTION
NAME	NO.		
COMP	1	I/O	Compensation pin
EN	3	I	Shutdown control input. Connect this pin to logic high level to enable the device
FB	2	I	Feedback pin
FREQ	7	I	Frequency select pin. The power switch operates at 650 kHz if FREQ is connected to GND and at 1.2 MHz if FREQ is connected to IN
IN	6	I	Input supply pin
PGND	4		Power ground
SS	8	O	Soft-start control pin. Connect a capacitor to this pin if soft-start needed. Open = no soft-start
SW	5	I	Switch pin

7 Specifications

7.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)(2)

	MIN	MAX	UNIT
Input voltage range IN	–0.3	7	V
Voltage range on pins EN, FB, SS, FREQ, COMP	–0.3	7	V
Voltage on pin SW	-0.3	20	V
Continuous power dissipation	See *Thermal Information*		
Operating junction temperature	–40	150	°C
Storage temperature	–65	150	°C

(1) Stresses beyond those listed under *absolute maximum ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *recommended operating conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability
(2) All voltage values are with respect to network ground terminal.

7.2 ESD Ratings

		VALUE	UNIT
$V_{(ESD)}$ Electrostatic discharge	Human body model (HBM), per ANSI/ESDA/JEDEC JS-001(1)	±2000	V
	Charged-device model (CDM), per JEDEC specification JESD22-C101(2)	±500	
	Machine model (MM)	±200	

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process. Manufacturing with less than 500-V HBM is possible with the necessary precautions. Pins listed as ±XXX V may actually have higher performance.
(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process. Manufacturing with less than 250-V CDM is possible with the necessary precautions. Pins listed as ±YYY V may actually have higher performance.

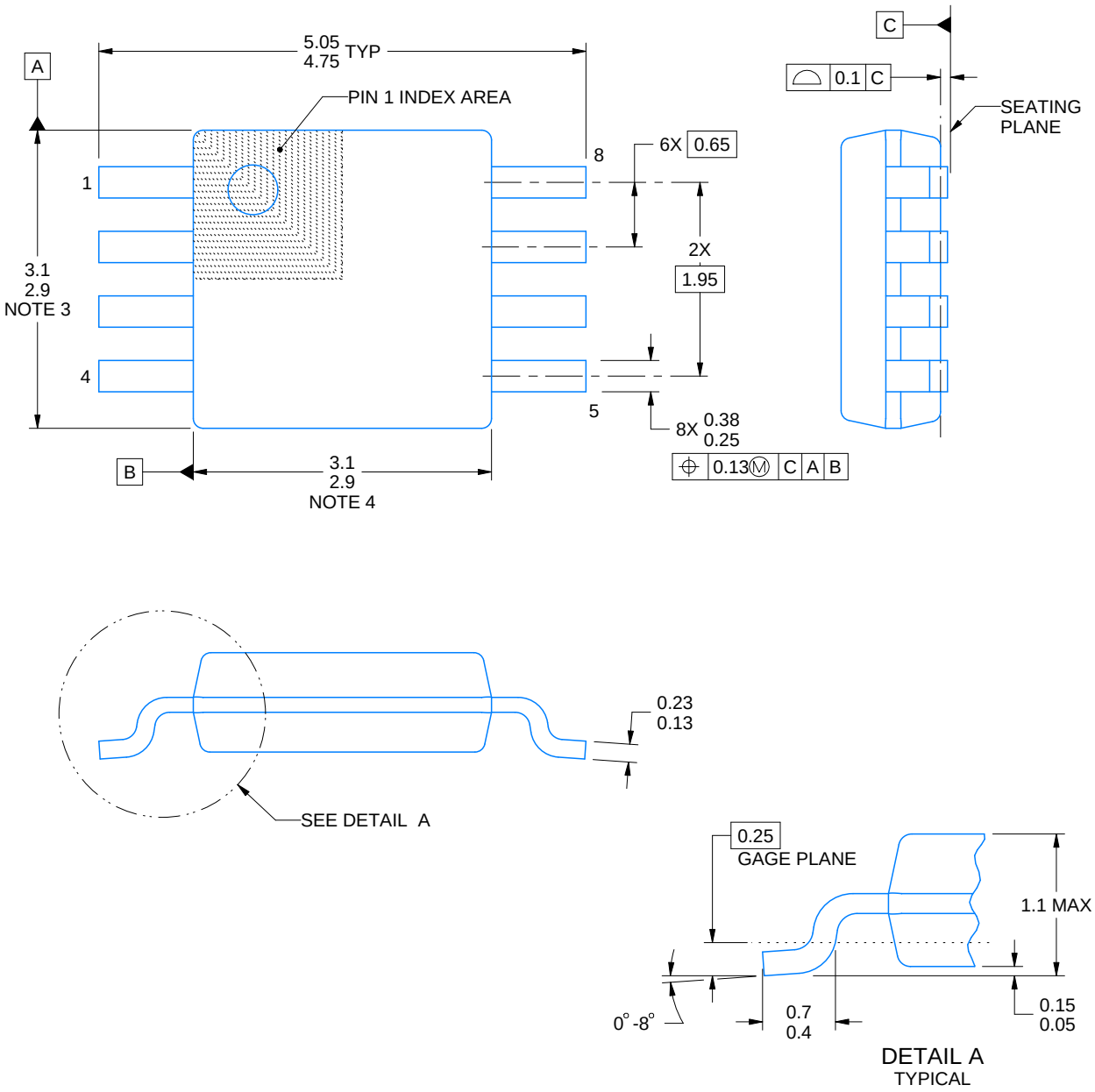
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PACKAGE OUTLINE

DGK0008A

VSSOP - 1.1 mm max height

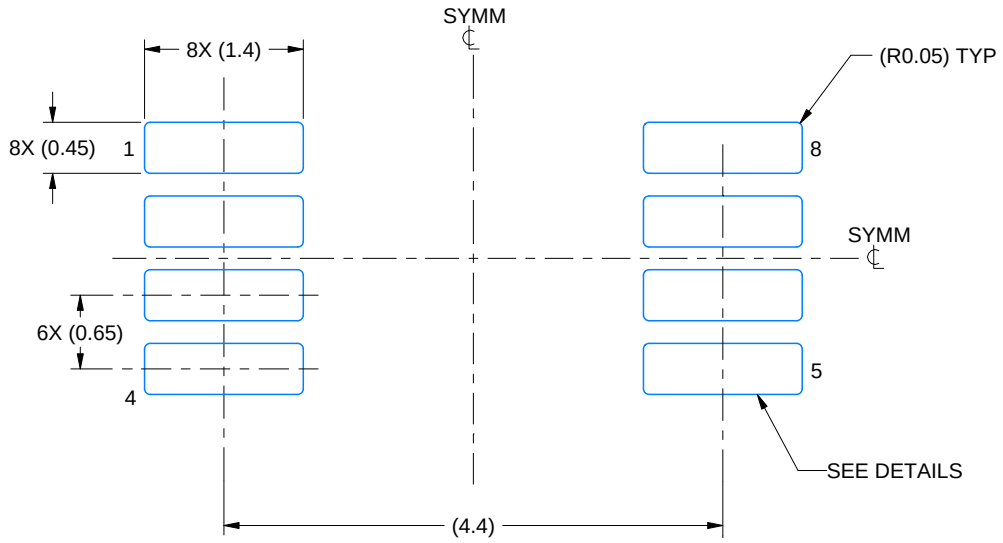
SMALL OUTLINE PACKAGE



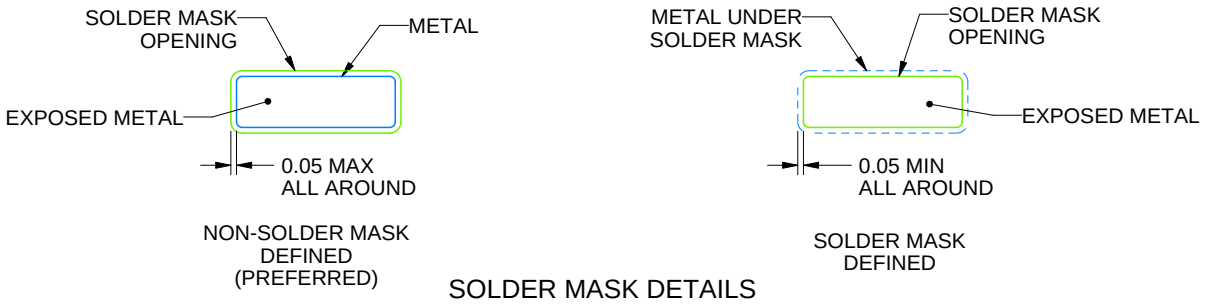
PowerPAD is a trademark of Texas Instruments.

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-187.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 15X



SOLDER MASK DETAILS

4214862/A 04/2023

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.
8. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.
9. Size of metal pad may vary due to creepage requirement.